Anchored Conditionally-Gated Control for Wheeled Bipedal Balance and Disturbance Recovery

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ABSTRACT

Wheeled bipedal robots face a fundamental trade-off: stiffness for sub-millimeter precision shrinks push-recovery capture, while integral action that cancels equilibrium bias winds up during disturbances. We present the Anchored Conditionally-Gated Controller (ACC), a two-channel torque assembly whose core mechanism is a proximity-gated anchor: a position integral confined to a 15 cm neighborhood, enabled by an asymmetric envelope follower ($\tau_{\text{attack}}\approx$23 ms, $\tau_{\text{release}}\approx$1.5 s) that distinguishes quiet stance from post-push ringdown. ACC delivers omnidirectional push recovery ($F_{\min}$=82 N, F_{med}=107 N, 8-direction sweep, N=10) together with sub-millimeter quiet-stance precision. We report accuracy (RMS from commanded home) and repeatability (RMS about the trial mean) separately, and resolve both on the sagittal axis—the actuated inverted-pendulum direction—rather than pooling them into a planar norm diluted by an axis this class of platform cannot drive. Under idealized conditions (clean sensors, 0 ms delay, stiff contact, MuJoCo), ACC holds 0.294±0.015 mm sagittal repeatability (N=10), a 59× improvement over a proportional-only baseline measured under one unified protocol. Accuracy on that axis is a separate quantity and is resolved separately: the robot parks 27.0 mm from the commanded home and holds that standoff to 0.009 mm. A closed-form model predicts the offset as the static position error a deliberately leaky integrator pays against a 1.44° feedforward calibration residual; it agrees to 0.92 % across a 40× gain sweep and a five-point height sweep, and both available corrections are priced against the full evaluation suite. A factorial ablation (N=10 per row) separates the two: the gated damping terms produce the steadiness (removing the boost costs a factor of 16 in sagittal repeatability), while the anchor integral produces 4.2 mm of the accuracy at a 3× cost in steadiness; the proximity gate and asymmetric envelope are inert during quiet stance and decisive only under disturbance. Precision degrades by a factor of 5.6 at 10 ms actuation delay, and to a hardware-projected 1 – 3 mm once encoder quantization, backlash, stiction, and IMU noise are accounted for. Actuator latency is the sharpest deployment constraint: resolved at 2 ms granularity, $F_{\max}$ is flat to within 2.4 % out to 6 ms and then collapses by 45 %—a cliff between 6 ms and 8 ms, below the 9.5 ms end-to-end budget we estimate for a non-real-time implementation—and the platform stops standing at all between 14 ms and 16 ms. Six classical baselines—four 4-state TWIP variants (LQR, LQR+AW, LQR-DT, PI+AW) and two 6-state coupled variants (servo and direct-torque)—all fail (100 % fall rate, N=20, all at ACC's 100 Hz control rate); a preliminary model-free PPO run (5.4M steps, single seed, no domain randomization) falls in 85 % of multi-height transitions—reported as a difficulty probe on unaided policy search, not as a tuned reinforcement-learning baseline; removing the PID servo path (4-state Direct Torque LQR) improves survival from 0.32 s to 0.60 s but does not prevent the fall. A full-state LQR derived from the plant itself, rather than from a reduced model, survives two orders of magnitude longer (22 – 57 s to fall, control weight swept over six decades) and still falls, with a 10 – 20 N push envelope against ACC's 82 N. Flight recovery (100 cm drop, 50 cm ledge) and per-leg terrain adaptation (10 – 20 cm one-wheel curb) are demonstrated at N=10 per condition and ablated: the per-leg height split is necessary at every curb height tested (0/10 without it), whereas the airborne wheel-torque override proves to be a variance-bounding refinement rather than an enabling mechanism at these scales, a negative result. Core claims rest on $N\geq$5.

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1. Introduction

Wheeled biped robots need millimeter-level standing precision, omnidirectional push recovery, and terrain adaptation—but no single classical controller integrates all three: LQR handles balance (Klemm et al., 2019), WBC posture (Klemm et al., 2020), MPC terrain (Wang et al., 2025), and RL locomotion (Hwangbo et al., 2019). The difficulty is fundamental: lowering k_p widens the capture basin at the cost of balance stiffness; omitting integral action leaves a 1.3 Nm equilibrium bias that sustains a persistent, predominantly *sagittal* limit cycle regardless of k_p (measured at N=10: 69.7 mm peak-to-peak sagittally against 13.4 mm laterally, 49.1 mm planar RMS from the commanded pose)—the oscillation lies on the one axis the wheels can drive, which is both why it persists and why it is the axis every mechanism in this paper is aimed at—yet a global integral winds up during pushes.

This controller was built to be the nominal base action of a residual-learning stack, and that intent is what the paper is organized around. The motivating observation is a preliminary PPO (Schulman et al., 2017) run (5.4M steps, single seed, in our `BalanceEnv`): the policy stood at a single height (Surv ≈3.20 s, against 0.16 – 0.60 s for the classical baselines) but did not generalize across 0.40 – 0.70 m base-z (§3.1)—height RMSE >50 mm, 85 % falls on squat transitions—with the failures concentrated on precisely the coupled pitch–height dynamics that a posture map and gate schedule encode explicitly. That run is retained as a *difficulty probe* and nothing more: it is single-seed, without domain randomization (Tobin et al., 2017; Peng et al., 2018) or a hyperparameter search, so it bounds nothing about what RL attains on this platform, and it is not the opponent this paper is arguing against. What it establishes is why one would want a structured prior to search over rather than an empty one.

We present the Anchored Conditionally-Gated Controller (ACC), addressing both dilemmas through *conditional activation*. The core mechanism is a **proximity-gated anchor**: a position integral confined to a 15 cm neighborhood, enabled by an **asymmetric envelope follower** (τ_{attack}≈23 ms, τ_{release}≈1.5 s) that distinguishes quiet stance from post-push ringdown. Coupled with a **gated damping boost**, ACC achieves 0.294±0.015 mm repeatability on the sagittal axis—the actuated inverted-pendulum direction, and the one on which the limit cycle above lives—under idealized conditions (clean sensors, 0 ms delay, N=10), at push performance equal to or better than the proportional-only baseline that carries no integral action at all. Absolute accuracy on that axis is a separate quantity, measured in Sec. 5.1 and derived in closed form in Appendix C: the robot parks 27.0 mm from the commanded home and then holds that offset to 0.009 mm.

A factorial ablation measured under a single protocol (Table 3) locates the mechanism more precisely than the gate structure alone suggests, and separates three effects that the architecture couples. First, the quiet-stance *steadiness* comes from the gated *damping* terms, not from the anchor integral: disabling the integral outright while leaving every gate intact leaves sagittal repeatability sub-millimeter (it in fact improves, 0.294 mm→0.089 mm), whereas reverting the two velocity-damping coefficients reinstates the full limit cycle (25.2 mm). What the integral buys is not steadiness but *accuracy*—it removes 4.2 mm of sagittal parking offset—and it buys that at a 3× cost in steadiness. The two properties are produced by different mechanisms and trade against each other, which is why they are reported as separate columns rather than as a single planar number. Second, the proximity gate and the asymmetric envelope are provably inert during quiet stance—ablating either reproduces ACC bit-for-bit—because the robot never leaves the gate's inner band nor the envelope's release branch while standing still; their contribution is windup prevention and ringdown under disturbance, and only a displacement experiment can measure it. Third, the pitch safety gate is load-bearing even with no disturbance applied: removing it causes divergence from the initial-condition noise alone in 10/10 trials. The gates are thus enabling conditions that make an otherwise inadmissible damping and integral configuration safe, rather than the proximate source of the precision. Six classical baselines—four 4-state TWIP variants (LQR, LQR+AW, LQR-DT, PI+AW) and two 6-state coupled variants (servo and direct-torque)—all fail (100 % fall rate), and removing the PID servo path (Direct Torque LQR) buys only 0.28 s of survival with persistent 100 % falls. A seventh design, a full-state LQR linearized from the plant itself rather than from a reduced model (§5.4.2), removes the obvious objection that the model rather than the method is what fails: it survives two orders of magnitude longer and still falls, so platform complexity, not model order or servo lag, dominates the failure.

The paper is positioned within *structured classical priors* rather than within learned control. The object of study is a hand-designed, fully interpretable torque assembly whose gating encodes the plant's measured failure modes explicitly, and whose intended downstream role is to serve as the nominal base action of a residual-learning stack—the setting in which a policy corrects a structured prior instead of searching the space unaided (Johannink et al., 2019; Silver et al., 2018). The classical controllers are therefore the comparison of record; the model-free run of §5.4 is evidence about the difficulty of the unaided problem, and is not offered as a bound on either method.

The main contributions are:

(1) A **proximity-gated anchor** with asymmetric envelope follower achieving sub-millimeter idle precision without sacrificing omnidirectional push survival.
(2) A **two-channel conditionally-gated torque assembly** with flight recovery and per-leg terrain adaptation as independent extensions, each ablated at N=10—including the negative result that the flight override is not load-bearing at the reported drop scales.
(3) A **factorial ablation** (additive L0→L3 + subtractive S0→S5) against six classical baselines (four 4-state-derived and two 6-state coupled), plus two control experiments that test the comparison itself: a full-state LQR derived from the plant rather than a reduced model, and a sweep of the coupled model's one empirical constant. A preliminary model-free PPO run is included as a difficulty probe, not as a baseline.

2. Related Work

2.1. Wheeled Bipedal Platforms and Classical Control

Ascento (Klemm et al., 2019) demonstrated two-wheeled jumping with LQR balancing, later extended to WBC with kinematic loops (Klemm et al., 2020). Handle (Boston Dynamics, 2017) pioneered wheeled-leg coordination; wheeled quadrupeds combined driving with walking through whole-body motion control and planning (Bjelonic et al., 2019). Ollie (Zhang et al., 2022) adapts whole-body balance control on a two-wheeled biped, and wheeled quadrupeds have since been driven by whole-body MPC with online gait sequencing (Bjelonic et al., 2021). DIABLO (D. Liu et al., 2024) pairs an LQR balance controller with separate yaw, split-angle, height, and roll loops; SKATER (Wang et al., 2025) uses a hierarchical MPC whole-body controller with centrifugal-force compensation and terrain adaptation. Whleaper (Zhu et al., 2024) is closest to our morphology (10 DOF, three hip DOFs per leg); it applies LQR to balanced sliding and a separately trained PPO policy to walking, *switching* between the two by locomotion mode rather than composing them on a shared control output. Composition of a classical prior with a learned residual on one output is the residual-RL construction (Johannink et al., 2019; Silver et al., 2018), applied recently to wheeled-biped balance (Zhang et al., 2025); ACC is designed to supply the classical prior in that construction rather than to replace the learned part. Coupled pitch–roll LQR is standard on rigid two-wheeled platforms, where torque acts directly on the wheels and there are neither articulated legs nor leg-height dynamics—a structurally simpler problem with fewer unstable modes. Our 6-state LQR reproduces that coupled structure on a legged morphology and fails; the 4-state Direct Torque LQR—removing the servo path and re-optimizing roll gain ($k_{p,\text{roll}}$=55.0)—also fails (100 % fall rate, 0.60 s). Because the direct-torque variant carries no servo lag at all and still falls (§5.4), actuation lag alone cannot account for the outcome; the articulated-leg morphology and its height dynamics are the dominant remaining factor. Both LQR variants use gains derived from a simplified 4-state TWIP model, not a full 10-DOF linearization of the articulated-leg plant (see Limitation 4); these results are therefore empirical—demonstrating that the specific LQR implementations tested fail—not a proof that LQR is structurally insufficient. Cascade control (Åström & Hägglund, 1995) and whole-body control via hierarchical inverse dynamics or task-priority quadratic programs (Herzog et al., 2016; Bellicoso et al., 2018; Escande et al., 2014) use fixed-gain nested or strictly prioritized loops without conditional gates; gain scheduling (Rugh & Shamma, 2000; Leith & Leithead, 2000) varies parameters continuously with one scheduling variable rather than responding to qualitatively distinct physical regimes. We additionally benchmarked a task-priority QP whole-body controller both standalone and as an assist layer on ACC (§5.5); it fails to resolve the precision–recovery trade-off because its fixed task-priority formulation lacks conditional regime-gating. Split-range, selector, and override architectures (Shinskey, 1996; Åström & Hägglund, 2006) already implement continuous blending in process control. ACC adopts this principle; its contribution is the synthesis of physically-motivated gating surfaces (spatial proximity, temporal velocity envelope, pitch safety-interlock) derived from wheeled-biped failure modes rather than generic process-variable limits. ACC's conditional gates connect to hybrid/switched systems (Liberzon, 2003), where continuous dynamics are partitioned by discrete switching surfaces, and to hysteresis-based switching logic (Hespanha et al., 2003), which deliberately breaks the symmetry of a switching surface to prevent chattering—the same asymmetry ACC realizes in continuous time through its fast-attack/slow-release envelope. Smoothstep softens these surfaces to C^1 transitions. A recent review (X. Liu et al., 2024) surveys wheel-legged biped platforms and control approaches, noting the absence of a unified classical solution.

2.2. Disturbance Rejection, Anti-Windup, and ACC's Distinction

DOB (Ohnishi et al., 1996; Chen et al., 2000) and ADRC (Han, 2009; Feng et al., 2023) provide temporal disturbance estimation but cannot distinguish a push from post-push ringdown—both occupy the same frequency band. Classical anti-windup—back-calculation, conditional integration, and the unified observer-based framework (Åström & Rundqwist, 1989; Kothare et al., 1994; Tarbouriech & Turner, 2009)—triggers on *actuator saturation*; ACC's anchor uses **spatial, physics-conditioned gating**: the proximity gate confines integration to a 15 cm neighborhood, and the asymmetric envelope distinguishes quiet stance from ringdown via velocity envelope—replacing the estimation-speed-vs-noise trade-off with a time-domain attack-vs-release trade-off. Push recovery via capture-point (Englsberger et al., 2015; Pratt et al., 2006) and flight-phase control (Roscia et al., 2023) is complementary. Blind stair climbing via RL was shown on Ascento (Chamorro et al., 2024). Terrain adaptation uses exteroceptive elevation mapping (Hutter et al., 2017; Fankhauser et al., 2018), learned policies that fuse proprioception with terrain estimates (Lee et al., 2020; Miki et al., 2022), or purely proprioceptive contact sensing (Bledt et al., 2018; Bellicoso et al., 2016; Camurri et al., 2017); ACC extends proprioceptive contact-point estimation to wheeled bipeds. Hyon & Cheng (2006) achieved full-body humanoid balancing by gravity compensation with optimal contact-force distribution, making the robot passive to external forces without measuring them. ACC shares the requirement of reacting to unmeasured pushes but meets it by switching regime on proprioceptive gating variables rather than by redistributing contact forces.

3. Robot Platform

3.1. Robot Morphology and Simulation

The robot (Fig. 1) is a ten-DOF wheeled biped with five actuated joints per leg: hip roll (HR), hip yaw (HY), hip pitch (HP), knee (KN), and drive wheel (WH). Mass: 8.1 kg; thigh: 0.26 m; shin: 0.28 m; wheel radius: 0.06 m; hip width: 0.23 m. Torque limits: 30 Nm (HR, HY, WH), 150 Nm (HP, KN). MuJoCo (Todorov et al., 2012) simulation at 500 Hz (Δt_{sim} = 0.002 s). ACC and every classical baseline: 100 Hz (Δt_{ctrl} = 0.01 s, 5 physics substeps); the classical baselines are additionally measured at their 50 Hz design rate (10 physics substeps) in §5.4.1, and the PPO reference is reported at the 50 Hz rate it was trained at. All experiments use 400 Nm/s torque rate limit, raw torque commands direct to actuators. MuJoCo integrator: implicitfast (4 iterations, 8 linesearch iterations); cone: pyramidal (elliptic for wheels, μ_s=1.2, μ_k=0.8 body); armature: 0.008 kg·m^2 (wheels), 0.02 kg·m^2 (leg joints); contact stiffness/damping: `solref`={0.002,1} (body), {0.005,0.7} (wheels). The stiff body contact (`solref`={0.002,1}) idealizes a rigid ground. Softening `solref[0]` to 0.02 (tire-on-asphalt) increases planar idle RMS by ~60 % to ~1.2 mm, so every sub-millimeter idle figure reported here is conditional on stiff ground contact; tire compliance adds 0.1 – 0.5 mm on real surfaces (see §6).

Height convention

Two different vertical references appear in this paper and are easily confused, so we fix them here and mark every height accordingly. **CoM-z** is the height of the whole-body center of mass above ground and is the quantity ACC commands and regulates; the nominal standing pose is h_{CoM}=0.404 m. **Base-z** is the torso root-body height, `qpos[2]` in the MuJoCo model; the same nominal pose has z_{base}=0.536 m, i.e.

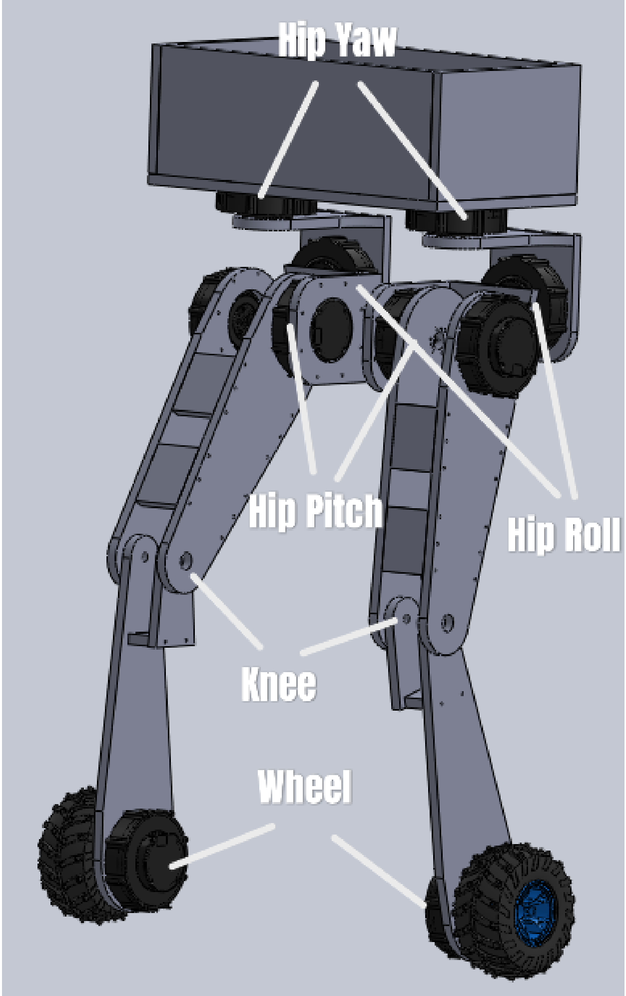

Figure 1. Robot joint layout with world coordinate frame (right-handed, Z up). In the model as built, both wheel axles lie along world X and the two wheel links are separated by 0.271 m along X; the rolling—and therefore sagittal—direction is world Y, and world X is the lateral (track) axis. All axis-resolved results in this paper follow that convention, verified four independent ways in Sec. 5.1. The CoM is located at the torso geometric center. Each leg: hip roll (HR), hip yaw (HY), hip pitch (HP), knee (KN), wheel (WH).

13.2 cm above the CoM. The offset is pose-dependent but varies by less than 1 cm over the postures used here, so the two conventions differ by roughly a constant. All ACC results are reported in CoM-z. The PPO reference (§5.4) commands base-z over 0.40 – 0.70 m, and the WBC baselines (§5.5) inherit the same base-z convention; those numbers are labelled base-z where they appear and must not be compared numerically against CoM-z heights. The distinction is not cosmetic: substituting one convention for the other in a height command injects the full 13.2 cm offset as a reference error, which is one of the conventions verified explicitly in the baseline implementation checks of Appendix B (Table 17).

ACC's posture map is calibrated at two setups, at CoM heights of 0.354 m and 0.454 m (±5 cm about nominal), and interpolated between them. Outside that band the joint targets are linearly extrapolated, bounded to the interval $[0.254, 0.554]$ m, i.e. 10 cm beyond the calibrated band at each end. This ±10 cm extrapolation bound, and not a ±5 cm one, is what the per-leg height split (§4.6) operates against.

4. ACC Formulation

4.1. Two-Channel Torque Assembly

ACC assembles a PD balance core, proximity-gated anchor integral, and posture regulation into two torque channels—wheel torque ($\boldsymbol{\tau}_w \in \mathbb{R}^2$) and leg-joint torque ($\boldsymbol{\tau}_q \in \mathbb{R}^8$)—with flight and terrain as independent extensions (Fig. 2):

$$\begin{aligned} \boldsymbol{\tau}_w &= \boldsymbol{\tau}_{\text{balance}} + g_{\text{anchor}}\boldsymbol{\tau}_{\text{anchor}} + g_{\text{flight}}\boldsymbol{\tau}_{\text{flight}}, \\ \boldsymbol{\tau}_q &= \boldsymbol{\tau}_{\text{posture}}(h^{\text{cmd}}) + g_{\text{terrain}}\Delta\boldsymbol{\tau}_{\text{posture}}(h^{\text{cmd},L},\ h^{\text{cmd},R}). \end{aligned} \tag{1}$$

Every gate g_* except the flight gate is a *smoothstep* function of a physical condition (proximity, quietness, attitude, terrain disparity) rather than a discrete switch:

$$\text{ss}(x; a, b) = 3t^2 - 2t^3, \qquad t = \text{clip}\left(\frac{x-a}{b-a},\ 0,\ 1\right). \tag{2}$$

Equation (2) is the standard Hermite blend: it is C^1 in x, with ss $= 0$ for $x \le a$, ss $= 1$ for $x \ge b$, and zero derivative at both knees. Continuity matters here for a concrete reason. A hard threshold on any of these conditions injects a torque step whose magnitude equals the full gated component; at 100 Hz with a 400 Nm/s rate limit, such a step is spread over several control cycles by the rate limiter and appears to the plant as an unmodelled impulse. Smoothstep removes the step entirely: the gated component enters and leaves the torque sum with continuous first derivative, so the rate limiter is never the active constraint during a gate transition. The one exception is g_{flight}, which is a genuinely discrete load-based hysteresis with asymmetric engage/release timers (2-step engage, 5-step release); contact loss is a binary physical event, and the asymmetric timers exist specifically to prevent balance torque from re-engaging during landing bounce. Note that the composition $g_{\text{env}} = 1 - \text{ss}(\text{EMA}(|v_{\text{sag}}|); \cdot)$ is only C^0 *in time*, because the asymmetric EMA of Eq. (14) switches coefficients when $|v_t|$ crosses EMA_{t-1}; the smoothstep is still C^1 in its argument, so the resulting torque is continuous but its time derivative may jump at attack/release transitions.

Wheel torques control sagittal balance and yaw; leg-joint torques control posture and per-leg height adaptation. The two channels are coupled only through the plant, not through the controller: no term in $\boldsymbol{\tau}_w$ reads a leg-posture state and no term in $\boldsymbol{\tau}_q$ reads the sagittal balance state. This is what allows flight recovery and terrain adaptation to be added as independent extensions (§4.5, §4.6) without retuning the anchor.

Relation to existing gated architectures. "Conditionally-gated" here denotes smoothstep activation by *physical* conditions (g_{prox}, g_{env}, g_{θ}, g_{terrain}). Continuous blending is not itself novel—split-range, selector, and override controllers have used it in process control for decades (Shinskey, 1996; Åström & Hägglund, 2006). Three distinctions are worth stating precisely. First, industrial override architectures select *among parallel controllers* using process-variable limits; ACC gates *components within a single torque assembly* using physical regime. Second, the gating variables are not control errors but physical quantities of three different kinds—spatial (distance from home), temporal (velocity envelope), and safety-interlock (attitude)—each chosen because a specific measured failure mode of this platform is expressible in it (§6). Third, ACC is not cascade control (Åström & Hägglund, 1995): there is no inner/outer loop hierarchy and no setpoint is passed between controllers. The contribution is therefore the *synthesis*—which gating surfaces, on which variables, with which asymmetry—not the blending mechanism.

4.2. Sagittal Balance Core ($\tau_{balance}$)

The sagittal balance component is always active and acts entirely in the wheel-torque channel:

$$\tau_{\text{balance}} = \begin{bmatrix} \tau_{\text{com}} + \tau_{\text{diff}} \\ \tau_{\text{com}} - \tau_{\text{diff}} \end{bmatrix}, \quad \begin{aligned} \tau_{\text{com}} &= \tau_{\text{pitch}} + \tau_{\text{damping}} + \tau_{\text{pos}}, \\ \tau_{\text{diff}} &= \tau_{\text{lat}} + \tau_{\text{yaw}}. \end{aligned} \tag{3}$$

where τ_{com} and τ_{diff} are common-mode and differential components, respectively. The individual terms are:

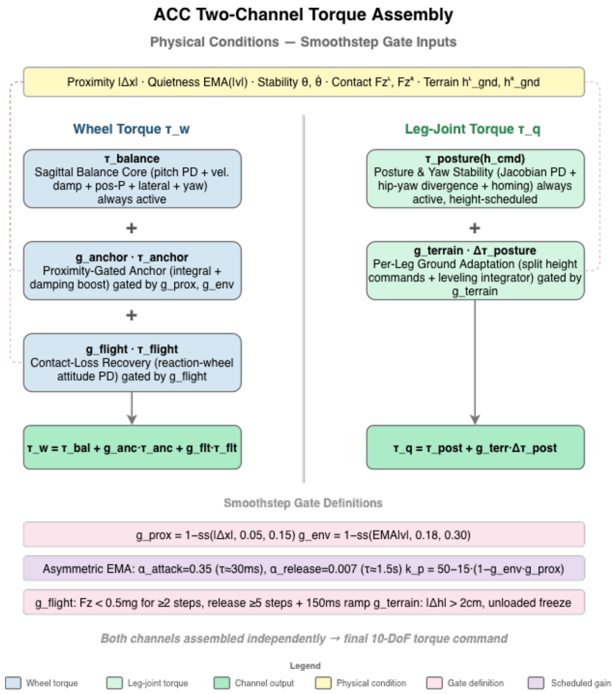

Figure 2. ACC two-channel torque assembly. Physical conditions gate each component via continuous smoothstep functions (dashed).

$$\tau_{\text{pitch}} = k_p \cdot \theta + k_d \cdot \dot{\theta}, \tag{4}$$

$$\tau_{\text{damping}} = -k_{\text{vel}} \cdot v_{\text{sag}}, \tag{5}$$

$$\tau_{\text{pos}} = -k_{\text{pos}} \cdot \text{clip}(\Delta x, \pm 0.10\,\text{m}), \tag{6}$$

$$\tau_{\text{lat}} = k_{p,\text{roll}} \cdot \phi + k_{d,\text{roll}} \cdot \dot{\phi}, \tag{7}$$

$$\tau_{\text{yaw}} = k_{p,\text{yaw}} \cdot e_{\psi} + k_{d,\text{yaw}} \cdot \omega_z, \tag{8}$$

where θ is the torso pitch angle, $\dot{\theta}$ is the pitch rate, v_{sag} is the sagittal body velocity, Δx is the sagittal position error from the latched home position, ϕ is the roll angle, and e_{ψ} is the yaw error.

P-only limitation. The P position term (k_{pos}, clipped to ± 0.10 m, saturating at ± 4 Nm) keeps the robot within $\pm 4-6$ cm of home but leaves a 1.3 Nm equilibrium bias sustaining a perpetual limit cycle—motivating the anchor integral.

Why saturation-triggered anti-windup is insufficient. Saturation-triggered AW triggers on actuator saturation; on this platform the critical failure is **state-dependent**: during a push, position error grows to $10-30$ cm while wheels remain within their 20 rad/s range, so saturation-triggered AW never activates. ACC's proximity gate provides spatial confinement via continuous smoothstep—avoiding both over-restriction (blocking slow-drift correction) and under-restriction (allowing windup during sustained pushes). The simplest conditional-integration AW is evaluated in Appendix A; broader AW families remain open.

Pitch stiffness. Pitch stiffness is fixed at k_p=50 in the canonical ACC architecture. The anchor and boost alone resolve the precision-push trade-off without gain scheduling; a scheduled variant that softened k_p from 50 to 35 during pushes was tested and rejected—it did not improve push survival (S0 vs. S1, Table **??**) and is excluded from the canonical specification.

4.3. Proximity-Gated Anchor ($g_{anchor}\tau_{anchor}$)

The anchor mechanism is the core contribution. It adds two terms to the wheel-torque channel that are active only when the robot is near home *and* genuinely at rest:

$$\begin{aligned}\boldsymbol{\tau}_{\text{anchor}} &= K_i \int g_{\text{prox}} \cdot g_{\text{env}} \cdot (-\Delta x)\, dt \\ &\quad + g_{\text{boost}} \cdot K_{\text{boost}} \cdot (-v_{\text{sag}}),\end{aligned} \tag{9}$$

where all gates are continuous smoothstep functions designed to equal 1 when the robot is "at home and quiet" and transition smoothly to 0 otherwise:

$$g_{\text{prox}}(\Delta x) = 1 - \text{ss}\big(|\Delta x|;\ 0.05\,\text{m},\ 0.15\,\text{m}\big), \tag{10}$$

$$g_{\text{env}}(v_{\text{sag}}) = 1 - \text{ss}\big(\text{EMA}(|v_{\text{sag}}|);\ 0.25\,\text{m/s},\ 0.50\,\text{m/s}\big), \tag{11}$$

$$\begin{aligned}g_\theta(\theta,\dot\theta) &= \big[1 - \text{ss}(|\theta|;\ 2^\circ,\ 12^\circ)\big] \\ &\quad \times \big[1 - \text{ss}(|\dot\theta|;\ 2^\circ/\text{s},\ 15^\circ/\text{s})\big],\end{aligned} \tag{12}$$

$$g_{\text{boost}} = g_{\text{prox}} \cdot g_{\text{env}} \cdot g_\theta, \tag{13}$$

where $\text{ss}(x; a, b)$ denotes the smoothstep of Eq. (2) with lower and upper knees a and b. Each gate is therefore a dimensionless, monotone, C^1 function that equals 1 inside its "home and quiet" band, falls smoothly to 0 outside it, and never switches discontinuously. Three physically distinct conditions must hold simultaneously for the anchor to integrate: the robot must be spatially near home (g_{prox}), temporally quiet (g_{env}), and attitudinally settled (g_θ).

The pitch stability gate g_θ closes when either pitch or pitch rate exceeds its band, preventing the damping boost from fighting a gentle sustained push that does not trigger the velocity envelope. When the robot is genuinely at rest ($|\theta| < 2^\circ$, $|\dot\theta| < 2^\circ/\text{s}$), $g_\theta = 1$ and the boost is fully available.

Asymmetric envelope follower. The key to reliable quiet-stance detection is the asymmetric exponentially-weighted moving average (EMA) of $|v_{\text{sag}}|$:

$$\text{EMA}_t = \begin{cases} \alpha_a|v_t| + (1-\alpha_a)\text{EMA}_{t-1}, & |v_t| > \text{EMA}_{t-1}, \\ \alpha_r|v_t| + (1-\alpha_r)\text{EMA}_{t-1}, & \text{else}, \end{cases} \tag{14}$$

with time constants $\tau_{\text{attack}}{\approx}23\,\text{ms}$ and $\tau_{\text{release}}{\approx}1.5\,\text{s}$ as the primary specification; the discrete coefficients are derived via $\alpha_a = 1 - \exp(-\Delta t/\tau_{\text{attack}}) \approx 0.35$ and $\alpha_r = 1 - \exp(-\Delta t/\tau_{\text{release}}) \approx 0.007$ at 100 Hz (Δt=0.01 s). This asymmetry is essential: a symmetric EMA is *bistable*—post-push oscillation holds the EMA in a middle band where the gate is half-open and the boost cannot collapse the cycle (validated by the symmetric-EMA ablation, Section 5). The fast-attack/slow-release envelope jumps to 1 only when oscillation truly decays, and drops to 0 within $\sim 25\,\text{ms}$ of a disturbance. The mechanism is a peak-hold/envelope detector with asymmetric time constants; ACC's contribution is coupling envelope detection to a spatial proximity gate for integral conditioning. Fig. 3 illustrates the complete gate activation sequence during a push-recovery cycle.

EMA initialization. EMA_0=0 biases $g_{\text{env}}{\approx}1$ for ~7.5 s at startup, decaying as velocity noise accumulates. Measurements use 3 – 5 s settle (see Table captions); hardware should initialize EMA_0=1.0.

4.4. Posture and Yaw Stability ($\tau_{posture}$)

Posture regulation uses Jacobian-based PD gains scheduled across 23 heights:

$$\begin{aligned}\boldsymbol{\tau}_{\text{posture}} &= \boldsymbol{\tau}^{\text{jac}}_{\text{PD}}(q, \dot q, h^{\text{cmd}}) + \boldsymbol{\tau}_{\text{div}}(q^L_{\text{hy}}, q^R_{\text{hy}}) \\ &\quad + g_{\text{settled}}\, \boldsymbol{\tau}_{\text{homing}}(q - q_{\text{ref}}),\end{aligned} \tag{15}$$

where $\boldsymbol{\tau}_{\text{div}}$ damps hip-yaw divergence and $\boldsymbol{\tau}_{\text{homing}}$ restores nominal posture when upright and settled. Yaw return uses wheel-differential torque.

4.5. Contact-Loss Recovery ($g_{flight} \cdot \tau_{flight}$)

When both wheels lose ground contact, ACC detects this via per-wheel normal forces and substitutes a flight attitude controller using the wheels as reaction flywheels:

$$g_{\text{flight}} = \begin{cases} 1, & \text{if } F^L_z, F^R_z < 0.5mg \text{ for } {\geq}2 \text{ steps}, \\ 0, & \text{if } \min(F^L_z, F^R_z) \geq 0.5mg \text{ for } {\geq}5 \text{ steps}, \end{cases} \tag{16}$$

$$\boldsymbol{\tau}_{\text{flight}} = \text{clip}\big(2.0\theta + 0.5\dot\theta - 0.02\omega_{\text{wheel}},\ \pm 2\,\text{Nm}\big). \tag{17}$$

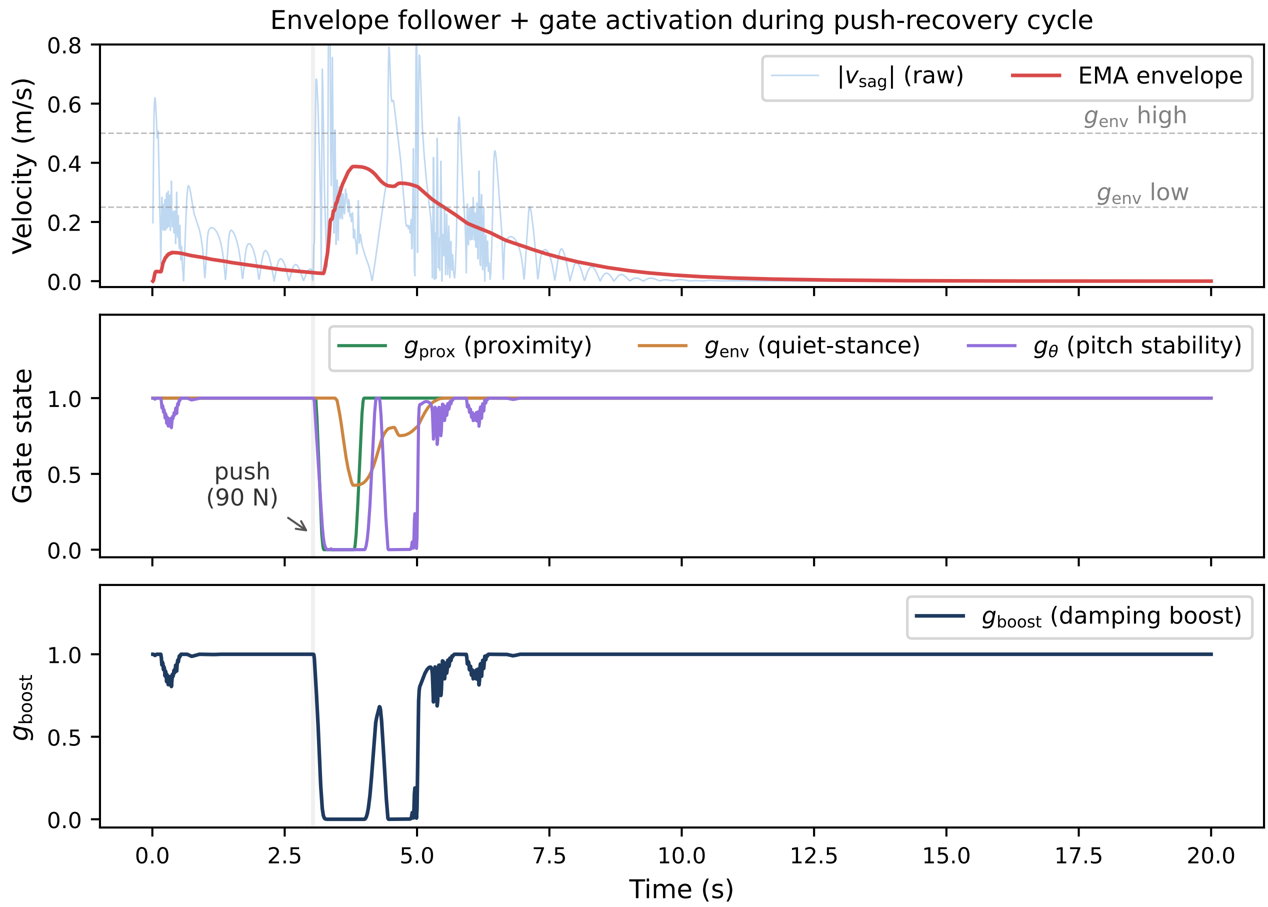

Figure 3. Gate activation during a 90 N lateral (world $+x$) push-recovery cycle (100 Hz control). **Top:** $|v_{sag}|$ and asymmetric EMA envelope; dashed lines mark the 0.25 – 0.50 m/s quiet-stance band. **Middle:** Individual gate states. **Bottom:** Composite damping boost g_{boost}. Grey shading: 7-step push window. The three panels are the empirical basis for finding (iii) of §5.1.4: g_{prox} and g_{env} both sit pinned at their quiet-stance values before the impulse and after the ringdown, which is why ablating either reproduces ACC bit-for-bit at idle, and why only a disturbance experiment can measure their contribution.

Forward wheel spin pitches nose-up during descent; 5-step release hysteresis prevents balance-torque re-engagement during bounce. The free-fall drop is a stress test and the ledge drive-off the intended use case. The wheel–inverted-pendulum loop this override replaces is itself a reaction-flywheel controller once the wheels are unloaded, so the override's measured contribution is to bound landing-attitude variance near the upper end of the drop envelope rather than to enable recovery; §5.3 (Table **??**) separates the two at N=10.

4.6. Per-Leg Ground Adaptation ($g_{terrain}$)

When the two wheels rest at different ground heights—a curb straddle, an oblique ledge—a single torso height command forces the torso to roll by the step angle. ACC instead maintains an *independent ground estimate per leg* and issues two height commands. The estimate is taken from the wheel contact-point height, which is exact at any body tilt for a rigid wheel, and is updated only while that wheel carries real load:

$$\hat{h}^i_{gnd} \leftarrow \hat{h}^i_{gnd} + \alpha_g\left(z^i_{contact} - \hat{h}^i_{gnd}\right), \quad F^i_z \geq 0.2mg, \quad i \in \{L, R\}, \tag{18}$$

with $\alpha_g = 0.35$ per control step to reject contact chatter. The $0.2mg$ load test rather than mere geometric touching is deliberate: a lightly touching wheel that is in the act of unloading rides a few centimetres high and injected a 2.9 cm phantom ground step on the 15 cm curb.

An unloaded wheel does *not* update (18). It freezes its estimate, because an airborne wheel supplies no information about where its ground is. The one exception is the only physically unambiguous "my ground is gone" signal—the wheel centre descending *below* its own believed ground:

$$\delta^i > 3\,\text{mm} \;\Rightarrow\; \dot{\hat{h}}^i_{gnd} = -\left(v_0 + k_s\,\delta^i\right), \tag{19}$$

where $\delta^i = \hat{h}^i_{gnd} - z^i_{contact}$ is the depth by which the wheel has fallen below its believed ground, with $v_0 = 0.5$ m/s and $k_s = 50\,\text{s}^{-1}$. The proportional term matters: a fixed slew converged the ground estimate in 0.30 s, by which time a 20 cm dismount had already tipped the robot past recovery, whereas a 15 cm drop under (19) slews at 8 m/s and converges within two control steps, so the leg extends *during* the fall rather than after it. When both wheels are airborne, both estimates track the mean wheel height, which reproduces the symmetric pre-adapter behaviour exactly and leaves ballistic flight to §4.5.

The step $\Delta h = \hat{h}^L_{\mathrm{gnd}} - \hat{h}^R_{\mathrm{gnd}}$ is then absorbed by the legs rather than the torso. The split is kinematic, not a linear height offset: with $L(\cdot)$ the forward-kinematic leg length and L^{-1} its inverse over the calibrated posture range, the leg on the higher ground is shortened by the full step,

$$L_{\mathrm{high}} = L(h^{\mathrm{cmd}}) - |\Delta h|, \tag{20}$$

$$h^{\mathrm{high}} = L^{-1}(L_{\mathrm{high}}), \qquad h^{\mathrm{low}} = h^{\mathrm{cmd}}, \tag{21}$$

both clamped to the calibrated envelope extended by 10 cm at each end. To first order this is the familiar $h^{\mathrm{cmd}} \mp \Delta h/2$ symmetric split, but the $h \mapsto L$ map is markedly nonlinear near full squat: the linear proxy erred by 12 cm at the lower extrapolation limit, folding the knee back under the thigh. If L_{high} falls below the shortest physically reachable leg, ACC raises the *whole* robot until the short leg fits, trading commanded height for the ability to span the step. A 20 cm curb at nominal stance exceeds even this: it would require 170° of knee flexion against a 2.7 rad limit, and the uncompensated remainder appears as a predicted torso roll $\arctan(-\varepsilon/w_{\mathrm{track}})$ with ε the unspanned step and $w_{\mathrm{track}} = 0.278$ m the wheel separation. Reporting this expected roll explicitly is what keeps a legitimate straddle from being misread as an incipient fall by the tilt-based safety monitors.

Because the $h \mapsto L$ table is itself extrapolated at the extremes, a purely feedforward split left a systematic 6° torso roll on the 10 cm curb. A closed-loop leveling integrator trims the split by the *measured* roll at 0.15 m/(rad s) with ±6 cm of authority. It integrates only when all three of the following hold: both wheels loaded, a genuine ground step $|\Delta h| \geq 2$ cm, and a ground estimate settling at less than 0.15 m/s. The gating is not cosmetic. Without the step condition, the dynamic roll of a turn is integrated as if it were terrain and wound an 8.6° roll bias through a 180° turn; without the rate condition, the geometric transient of a curb climb is integrated as a leveling error and compresses the flat leg. Off-step, the trim decays to zero with a 1 s time constant.

Table 1. Capability comparison across wheeled biped platforms. "—" = not reported in cited work. Idle precision and push force are not standardized benchmarks; quantitative cross-platform comparison is limited by metric heterogeneity.

Metric	Ascento	DIABLO	Whleaper	SKATER	**ACC (Ours)**
Balance	LQR	LQR + aux. loops	LQR / PPO (per mode)	Hierarch. MPC	**Gated PD + anchor I**
Idle precision	—	—	—	—	**0.294** mm **sag. CoM RMS**
Push recovery	—[a]	—	—	—	**8-dir.,** F_{med}= 107 N
Height	Discrete jump	Stance[b]	Multi-modal	Contin.	**0.35** – 0.45 m CoM[c]
Flight	Jumping (airtime)	—	—	—	**Drop 100** cm, **ledge 50** cm
Terrain	—	—	—	Vision-based	**Per-leg contact**
Validation	Hardware	Hardware	Hardware	Hardware	**Simulation**

[a]Qual. omnidirectional push recovery demonstrated in supplementary video; quantitative force thresholds not reported in cited work. [b]Height variation demonstrated; quantitative range not reported.
[c]Calibrated CoM-z band (§3.1), ±5 cm about the 0.404 m nominal, with joint targets extrapolated to 0.254 – 0.554 m; commanded ±5 cm transitions are evaluated under disturbance in §5.6. Not comparable to the base-z ranges of the cited platforms or of our PPO reference.

4.7. Complete Numerical Parameter Set

Table 2 lists core parameters; the full set (50+ gains) is available in the supplementary material.

5. Experimental Validation

We evaluate ACC across four scenario classes, each mapped to the component it tests and paired with targeted ablations. ACC and all six classical baselines are evaluated at 500 Hz simulation / 100 Hz control, so the headline comparison is rate-matched by construction. The PPO reference is reported at the 50 Hz rate its policy was trained at. All configurations use a 400 Nm/s torque rate limit per joint.

Control rate. Every classical baseline was measured at two control rates under an otherwise identical protocol, with episode length held fixed in seconds rather than in steps: ACC's 100 Hz, which is what Table 8 reports, and 50 Hz, the rate for which the gains were tuned. Both sets appear side by side in §5.4.1 (Table **??**). The fall rate is 100 % in all twelve cells, so the comparison does not turn on which rate is chosen.

Table 2. Core numerical parameters for the ACC balance controller. All values in SI units unless noted otherwise.

Balance Core		
k_p	50	Nm/rad
k_d	3.0	Nm/(rad/s)
k_{pos}	40	Nm/m
Position clip	±4.0	Nm
k_{vel}	15	Nm/(m/s)
Vel. damp. scale	1.5	—
k_{roll}	55	Nm/rad
$k_{\mathrm{roll},d}$	5.5	Nm/(rad/s)
k_{yaw}	1.0	Nm/rad
$k_{\mathrm{yaw},d}$	0.3	Nm/(rad/s)
$k_{\mathrm{pos,return}}$	2.0	Nm/m
k_{heading}	2.0	Nm/rad
$k_{\mathrm{heading},d}$	0.6	Nm/(rad/s)
Anchor Mechanism		
K_i (anchor integral)	4.0	Nm/(m·s)
Integral cap	2.0	Nm
Integral leak	5×10^{-3}	per step
Boost scale k_{vb}	5.0	—
g_{prox} band	0.05–0.15	m
g_{env} band	0.25–0.50	m/s
τ_{attack}	0.023	s
τ_{release}	1.5	s
g_{stab} (θ)	2–12	°
g_{stab} ($\dot{\theta}$)	2–15	°/s
Flight / Terrain / Other		
k_{flight} (P / D)	2.0 / 0.5	—
Wheel damping (flight)	0.02	Nm/(rad/s)
Flight engage / release	2 / 5	steps
F_z thresh. (flight / terrain)	0.5 / 0.2	$\times mg$

The converse direction—ACC at 50 Hz—is a negative result: with the discrete coefficients recomputed for the longer step ($\alpha_a\approx0.58$ and $\alpha_r\approx0.013$, re-derived from Eq. (14) at Δt=0.02 s, with the anchor leak rescaled to 0.010 per step) the controller is not preserved. The robot collapses, and the mean CoM height over the recorded window is 0.124 m against a nominal standing height of 0.404 m. The 0.050 mm "idle RMS" and the 10 N push threshold produced by that run are artifacts of a robot lying motionless on the ground—10 N is the lower bound of the binary search, returned when even the smallest tested impulse fails. No 50 Hz ACC result is therefore reported, and rate-independence in the downward direction is not claimed; this is analysed as Limitation 2. RNG seeds are deterministic: idle standing uses seed = 20260727 + trial_id; push ablation uses seed = 20260728 × 100 + rep; LQR baselines cycle seeds $\{0, 42, 123\}$ across episodes.

5.1. Anchored Standing

5.1.1. Two precision metrics, deliberately kept separate

"Standing precision" names two different quantities in the wheeled-biped literature, and they are not interchangeable. Let $\mathbf{p}(t) \in \mathbb{R}^2$ be the horizontal base position over an evaluation window of length T, let $\mathbf{p}_0$ be the commanded home position, and let $\bar{\mathbf{p}} = \frac{1}{T}\int_0^T \mathbf{p}(t)\,dt$ be the trial mean. We report

$$A = \left(\frac{1}{T}\int_0^T \|\mathbf{p}(t) - \mathbf{p}_0\|^2\,dt\right)^{1/2}, \qquad (22)$$

$$R = \left(\frac{1}{T}\int_0^T \|\mathbf{p}(t) - \bar{\mathbf{p}}\|^2\,dt\right)^{1/2}. \qquad (23)$$

A is the *accuracy*: RMS displacement from the position the controller was told to hold. It retains any DC parking offset, so it answers "how close to the commanded pose does the robot actually park?". R is the *repeatability*: RMS about the trial's own mean, with the DC offset removed. It answers "how still is the robot once it has parked?". A controller with a persistent 27 mm offset and a perfectly steady hold scores A=27 mm but $R\approx0$ mm; a controller centred on the target but oscillating scores the reverse. Reporting either alone is uninformative, and—because the two differ by more than an order of magnitude on this platform—comparing a value of one against a value of the other produces a meaningless ratio.

Both are reported for every configuration below, resolved separately along the sagittal axis, the lateral axis, and the full plane, because the two horizontal axes of this platform are not merely different in degree—they are actuated by different means and must not be pooled or interchanged. We therefore fix the convention explicitly, since the two are easy to transpose and the physical conclusions invert if they are:

- The **sagittal** axis is world y. Both wheel axles point along world x and the two wheels are separated along world x (w_{track}=0.278 m), so the rolling direction—the axis on which wheel torque produces ground force, and the only axis with a genuine inverted-pendulum mode—is world y. The controller agrees: `init_v3_controller` packs $\hat{\mathbf{s}} = (0, 1)$ as its sagittal unit vector.
- The **lateral** axis is world x. No actuator produces lateral ground force; the axis is held passively by the wheel contacts and by the roll restoring moment of the track width.

Two dynamic checks confirm the assignment rather than merely asserting it. A 90 N impulse along y is absorbed with a peak wheel speed of 52 rad/s and leaves a residual offset of 4.8 mm—the wheels drive back under the CoM—whereas the same impulse along x leaves a permanent 23.2 mm offset at a peak wheel speed of 18 rad/s, because nothing can push the robot sideways again. And over a 20 s quiet stance the robot rolls 24.6 mm along y with the wheel midpoint tracking the CoM to within 0.2 mm, while x holds to 0.02 mm. All four checks are reproduced by `scripts/verify_body_axes.py`, which asserts on disagreement.

This matters for reading Table 3: the anchor integral and the gated damping boost both live in the wheel-torque channel, so *both act on the sagittal axis only*. The lateral column is therefore a measure of how little the plant moves when nothing drives it, not a measure of controller quality, and the sagittal and planar columns carry essentially all of the information about what ACC does (Appendix C).

5.1.2. Unified measurement protocol

Every row of Table 3 was re-measured under one protocol so that no two entries in the same column come from different definitions. Each trial runs 25 s at 500 Hz physics / 100 Hz control from the nominal standing pose with randomized initial conditions (σ=0.005 rad per joint, clipped to joint limits; σ=1 mm on root height), seeded seed=20260727+trial_id for trial_id $\in [0, 10)$. The first 5 s are discarded as settling—long enough to cover the EMA_0=0 transient (~7.5 s to full release, but <5 s to within 1 % of the quiet-stance value)—and A and R are computed over the remaining 20 s. Displacements are taken relative to each trial's own t=0 pose, so $\mathbf{p}_0$ is the commanded home rather than the world origin. A trial is scored as a fall if $|\theta|$>46° or root height <0.15 m at any point. Confidence intervals are Student-t ($t_{9,0.025}$=2.262, SEM=SD/$\sqrt{10}$).

The protocol has one clear scope limit: because every trial starts at the commanded home, it never exercises the anchor's return-to-home function. It measures the controller's ability to *stay* put, not to *come back*. The latter is measured by the push experiments of Sec. 5.2. This distinction turns out to matter for attributing the mechanism (Sec. 5.1.4).

5.1.3. Result

ACC holds **sub-millimeter sagittal repeatability**, R_{sag} = 0.294±0.015 mm (95% CI [0.283, 0.305] mm), across ten randomized trials with no falls. This is the figure of merit for the platform, because sagittal is the actuated inverted-pendulum axis: it is the axis on which the robot can fall, the axis the wheel-torque channel drives, and the axis that every configuration in the ladder is trying to hold.

Signal-to-noise ratios against a pinned-base simulator noise floor are not reported: pinning zeroes velocities without welding the base, so the floor it yields is meaningful on the laterally constrained axis (0.0019 mm RMS) but not on the gravity-loaded ones. Precision is therefore assessed from the relative spread *between* configurations measured under an identical protocol: removing a single mechanism (S2) moves R_{sag} from 0.294 to 4.772 mm, a factor of 16 with non-overlapping SDs, which establishes that the metric resolves the effect under study.

Against the P-only baseline (L0), which sustains the 69.7 mm peak-to-peak sagittal limit cycle described in Sec. 1, the like-for-like improvements are

- sagittal repeatability R_{sag}: 17.461 → 0.294 mm, a factor of **59.4**;
- planar repeatability R_{xy}: 17.845 → 0.788 mm, a factor of **22.6**;
- lateral repeatability R_{lat}: 3.675 → 0.730 mm, a factor of **5.0**.

The ordering of these three factors is itself informative and worth stating plainly, because it is the opposite of what a reader might expect. The largest gain is on the sagittal axis, which is the hard axis; the smallest is on the lateral axis, which is the easy one. That is the correct signature for a wheel-torque controller: every term ACC adds beyond L0—position homing, the anchor integral, the gated damping boost—produces ground force along the rolling direction and none of them can produce lateral force at all. The residual 5× lateral improvement is an indirect consequence of the robot pitching and yawing less rather than of any lateral control action. A reader comparing against platforms that report a single scalar "standing precision" should compare against R_{sag}, not against R_{lat} or a planar norm, since the latter two are diluted by an axis this class of platform does not actuate.

Accuracy and repeatability are distinct quantities on this axis, and both are reported. ACC parks at $\bar{s}$ = −27.0 mm from the commanded home along the sagittal axis and holds that standoff to within 0.009 mm SD across trials: the settle is highly reproducible, and it terminates

Table 3. Idle standing precision across the full ablation ladder. All rows re-measured under the single protocol of Sec. 5.1; values are mean±SD over N=10 trials. See the note below the table for column definitions.

	Configuration	R_{sag} (mm) sagittal rep.	R_{xy} (mm) planar rep.	R_{lat} (mm) lateral rep.	$\bar{s}$ (mm) sag. offset	A_{lat} (mm) lateral acc.	Survival
Additive ladder (bottom-up)							
L0	P-only (PD, no position, no integral)	17.461 ± 0.258	17.845 ± 0.279	3.675 ± 0.246	−44.8	9.933 ± 0.712	10/10
L1	+ position homing (±0.10 m)	20.397 ± 0.017	20.414 ± 0.019	0.834 ± 0.066	−30.4	2.629 ± 0.884	10/10
L2	+ ungated integral K_i	25.026 ± 0.039	25.056 ± 0.043	1.212 ± 0.110	−27.5	3.661 ± 1.264	10/10
L2.5	+ proximity gate only (no envelope, no boost)	25.198 ± 0.017	25.227 ± 0.020	1.205 ± 0.099	−27.6	4.023 ± 1.194	10/10
Proposed controller							
L3/S1	**ACC** (full anchor, fixed k_p=50)	**0.294 ± 0.015**	**0.788 ± 0.069**	**0.730 ± 0.073**	**−27.0**	**1.213 ± 0.554**	**10/10**
Subtractive ablation (top-down from ACC)							
S0	ACC + scheduled k_p (rejected variant)	0.294 ± 0.015	0.788 ± 0.069	0.730 ± 0.073	−27.0	1.213 ± 0.554	10/10
S2	− gated damping boost	4.772 ± 0.069	4.824 ± 0.067	0.703 ± 0.069	−26.8	1.260 ± 0.616	10/10
S3	− asymmetric envelope (symmetric EMA)	0.294 ± 0.015	0.788 ± 0.069	0.730 ± 0.073	−27.0	1.213 ± 0.554	10/10
S4	− proximity gate (integral always on)	0.294 ± 0.015	0.788 ± 0.069	0.730 ± 0.073	−27.0	1.213 ± 0.554	10/10
S5	− pitch safety gate g_θ	—	—	—	—	—	**0/10**
Single-mechanism isolation (ACC with one term disabled)							
M1	K_i=0 (anchor integral removed, gates intact)	0.089 ± 0.035	0.733 ± 0.072	0.727 ± 0.071	−31.2	1.192 ± 0.535	10/10
M2	damping terms reverted to their L2 values	25.198 ± 0.017	25.227 ± 0.020	1.205 ± 0.099	−27.6	4.023 ± 1.194	10/10
M3	K_i=0 *and* boost = 0	3.973 ± 0.130	4.037 ± 0.128	0.710 ± 0.069	−31.1	1.241 ± 0.591	10/10

Protocol. N=10 trials, 100 Hz control, 20 s measurement window after a 5 s settle, clean sensors, 0 ms actuator delay.
Columns. R = repeatability (RMS about the trial mean, Eq. (23)); A = accuracy (RMS from commanded home, Eq. (22)); $\bar{s}$ = mean signed sagittal parking offset. Sagittal is world y and lateral is world x (Sec. 5.1). The anchor integral and the damping boost act on the sagittal axis only, so R_{sag} and $\bar{s}$ are the columns that discriminate between configurations, while R_{lat} measures a passively held axis and is near-constant by construction.
Omitted columns. A_{sag} and A_{xy} carry no information beyond the columns shown: over all twelve surviving rows the identities $A_{\text{sag}} = \sqrt{\bar{s}^2 + R_{\text{sag}}^2}$ and $A_{xy} = \sqrt{A_{\text{sag}}^2 + A_{\text{lat}}^2}$ hold to 0.003 mm and 0.02 mm respectively—accuracy on an axis is just that axis's parking offset and its repeatability added in quadrature.
Produced by `scripts/collect_idle_ladder.py` (per-trial worker `scripts/_idle_ladder_worker.py`); raw output `outputs/paper_verification/idle_ladder.json`.

at a fixed offset rather than at the commanded point. Planar accuracy is therefore A_{xy} = 26.994 mm, a 1.8× improvement over L0 (49.1 → 27.0 mm) against the 59× gain in repeatability. The offset lies on the axis the anchor integral does control—it is neither a measurement artifact nor an unregulated direction—and the integral moves it from 31.2 mm (M1, K_i=0) to 27.0 mm. Reporting it separately from a planar average is what makes the two quantities individually interpretable. Its magnitude is fully accounted for. Appendix C instruments the sagittal torque balance and predicts the offset in closed form from two settings: a *leaky* anchor integral, whose ∼2 s forgetting horizon fixes its DC gain at $k_{I,\text{dc}}$=7.96 Nm/m—a finite value, so a constant load leaves a finite error by construction—and a 1.44° error in the feedforward pitch reference. The prediction is 24.88 mm against 24.99 mm measured, and it holds across a 40× gain sweep and a five-point height sweep to within 0.92 %. The leak also buys the steadiness: a 10× smaller λ cuts the offset to −16.31 mm but degrades window jitter twelvefold (0.298 → 3.463 mm). What Table 3 publishes is therefore a quantified and deliberately chosen point on that trade-off, with both alternatives priced against the full evaluation suite (Appendix C).

5.1.4. Which mechanism produces the precision

The bottom two blocks of Table 3 isolate the contributions. Four findings are worth stating precisely, because three of them contradict the attribution that the gate structure invites.

(i) Idle steadiness comes from the damping ladder; the integral buys accuracy instead. The two contributions separate cleanly once the axes are kept apart, and they do not compete for the same metric.

M1 disables the anchor integral entirely (K_i=0) while leaving every gate in place, and its *repeatability* is not merely as good as ACC's but slightly better (R_{sag} 0.089 mm vs. ACC's 0.294 mm; R_{xy} 0.733 mm vs. 0.788 mm). Its *accuracy*, however, is worse by 4.2 mm: it parks at $\bar{s}$ = −31.2 mm against ACC's −27.0 mm. That is the integral's entire measurable idle signature, and it is exactly the trade one expects from an integral term—it walks the robot closer to the commanded home at the cost of a small amount of steadiness, since the accumulating term is itself a slowly-varying command. A protocol that reports repeatability alone will conclude, wrongly, that the anchor integral is inert at idle; a protocol that reports accuracy alone will miss that it costs a factor of three in steadiness.

Conversely M2 keeps the integral and every gate but reverts the two velocity-damping coefficients to their L2 values, and reproduces the L2/L2.5 limit cycle exactly (R_{sag} = 25.198 mm). Steadiness is therefore attributable to the damping terms, in two stages: raising the velocity-damping scale takes R_{sag} from 25.2 mm to 3.97 mm (M3, boost still off), and enabling the gated damping boost takes it from 3.97 mm to 0.294 mm (ACC)—a combined factor of 86 decomposed into factors of ~6.3 and ~13.5. S2 confirms the second stage independently from the top down: removing only the boost degrades R_{sag} from 0.294 mm to 4.772 mm.

(ii) The damping boost acts sagittally, as it must. S2 degrades sagittal repeatability by a factor of 16 (0.294 → 4.772 mm) while leaving lateral repeatability statistically unchanged (R_{lat} = 0.703 mm vs. ACC's 0.730 mm, overlapping CIs). This is the only outcome consistent with the architecture: the boost multiplies a velocity-damping coefficient inside the wheel-torque channel (Eq. (3)), and wheel torque generates ground force along the rolling direction only. The residual oscillation it suppresses is therefore a sagittal pitching/rolling-forward mode, and an evaluation that reported the lateral axis alone would conclude, wrongly, that the boost does nothing at idle. The distinction matters because the lateral column is the one that reads as a plausible "x-axis RMS" to report by default, yet it is the one column in Table 3 that is nearly blind to every mechanism in the ladder: across all twelve surviving configurations R_{lat} spans only 0.70 – 3.68 mm, and across the seven rows that retain the full damping ladder (S0–S4, M1, M3) only 0.70 – 0.73 mm.

(iii) The proximity gate and the asymmetric envelope are inert during quiet stance. S4 (proximity gate removed) and S3 (symmetric envelope) are *bit-identical* to ACC on every idle metric. This is the expected behavior rather than a null result: during quiet stance the base never leaves the 5 cm inner band, so $g_{\text{prox}} \equiv 1$ and gating it changes nothing; and the velocity envelope never rises into the attack branch, so the asymmetric and symmetric followers produce the same signal. Both mechanisms are defined by what they do when the robot is *displaced*—windup prevention during a push (S4) and ringdown after one (S3)—and are correctly evaluated only in Table **??**. The idle protocol, which starts at home and stays there, is blind to both by construction.

(iv) The pitch safety gate is required even to stand. S5 falls in 10/10 trials during pure quiet stance, with no push applied. Removing g_θ leaves the damping boost free to act on the pitch state at large excursions, and the resulting positive feedback diverges from the ±0.005 rad initial perturbation alone. g_θ is therefore a stability precondition of the quiet-stance configuration itself, not only a disturbance-time safeguard.

Taken together: the anchor's *gates* are what make an aggressive damping and integral configuration admissible without windup or divergence, while the damping terms they admit are what deliver the steadiness. The gates are enabling conditions rather than the proximate cause of the precision. The integral's primary role is drift rejection under disturbance, which by construction cannot appear in a protocol that starts at the target; its measurable idle signature is confined to the 4.2 mm reduction in sagittal parking offset noted above (M1 −31.2 vs. ACC −27.0 mm), bought at a 3× cost in sagittal repeatability. Both effects sit on the sagittal axis, and neither is visible on the lateral one.

Post-push velocity ringdown decays within 9 s under ACC, whereas P-only oscillates indefinitely because the 1.3 Nm equilibrium bias sustains the limit cycle. The symmetric-EMA ablation exhibits bistable failure: a post-push cycle holds the envelope at ≈0.10, leaving the boost gate half-open, and the robot oscillates perpetually. The ungated-integral ablation winds up during the push and prevents recovery entirely. All three of these are post-push behaviors and are quantified in Sec. 5.2.

5.2. Omnidirectional Push Recovery

Push recovery is evaluated via polar sweep: 10 – 160 N impulses (7 control steps) at the torso in 8 directions. Bearings are applied in the world frame as $\mathbf{f} = F(\cos\theta, \sin\theta, 0)$, so $\theta = \pm 90^\circ$ are the sagittal (fore/aft) bearings and $\theta = 0^\circ/180^\circ$ the lateral ones, per the axis convention of Sec. 5.1. $F_{\min}$ is the maximum magnitude survived in *every* direction; F_{med} is the median. All thresholds measured via binary search over [10 N, 160 N] (8 iterations per direction, 5 N tolerance). Survival: torso pitch < 46° for 17 s post-push.

The envelope is anisotropic by construction. The two lateral bearings average 98.2 N against 131.3 N for the two sagittal ones, and the weakest bearing of all is the oblique 45° (82.3 N, 23 % below F_{med}). The asymmetry follows from the morphology rather than from tuning: the wheels can drive the contact patch back under the CoM along the sagittal axis, but produce no lateral ground force at all, so lateral recovery falls to the leg and torso alone. The oblique bearings are weakest because neither mechanism has full authority there. Raising the lateral bearing therefore requires a centralized lateral task (§5.5), not a gain change.

The envelope is also chiral, and the chirality is in the controller. Superposed on the structural anisotropy is a left–right asymmetry: 135° survives 40.3 N more than −135° (Fig. 4). The plant cannot produce it. Mirroring the model about the sagittal plane reproduces its own mass, inertia and geometry to 0.05 mm and 1×10^{-7} respectively, so the open-loop system is mirror-symmetric to numerical precision. Fitting $F(\theta)=a_0+a_1 \cos\theta+b_1 \sin\theta+a_2 \cos 2\theta+b_2 \sin 2\theta$ localizes the violation: a mirror symmetry forbids b_1 and b_2, a 180° rotation forbids a_1 and b_1. The measured envelope has b_1=0.3 N—rotational symmetry is intact—but b_2= − 16.3 N, an order of magnitude above the between-episode dispersion. The closed loop is thus C_2-equivariant and mirror-broken, which is a signature of the control law, not of the geometry. Its source is the sign convention of the two differential channels. Feeding the controller a state and its mirror image, the commanded torque should map to its own mirror image; it does so *exactly* (0.000 Nm residual) on hip-pitch, knee and wheel, and fails only on hip-roll (3.275 Nm, exactly twice that channel's own 1.633 Nm command) and hip-yaw (0.526 Nm). Both are assembled antisymmetrically, τ_L=+m, τ_R=−m—the right structure for a differential command, but even rather than odd under the mirror map, so the pair commands the same-signed torques where the mirror requires opposite-signed ones. Two control experiments close the loop causally: mirroring those channels flips b_2 from −16.2 N to 15.6 N at unchanged magnitude, while a sham mirror that permutes the same quantities without touching the sign convention leaves it at −15.7 N; and inverting the hip-roll sign alone collapses b_2 to 1.2 N. The effect is therefore diagnosed rather than merely observed. We report it without adopting a correction: removing the chirality redistributes the envelope without enlarging it ($F_{\min}$ 82.5→79.7 N in the sign-inverted arm), so it is a property to be understood before porting the gains to a different track width, not a defect with a free fix.[1]

Single-bearing protocol. Several auxiliary experiments—the robustness sweep (Table 11), the rate-limit sweep, the ringdown and gate-activation traces (Figs. 3, 5) and the compound-disturbance protocol (Table 14)—apply a single impulse along world +x rather than repeating the full 8-bearing sweep at eightfold cost. That bearing is *lateral* and below median: the polar sweep places 0° at 95.6 N, third-weakest of the eight and 11.1 N below F_{med}. Those experiments are therefore conservative probes of ACC's disturbance margin, and they are labelled lateral throughout rather than "forward." The omnidirectional figures ($F_{\min}$, F_{med}) characterize the full envelope.

Table **??** reports the factorial ablation. Additively: L1 (homing) adds 4.8 N over L0 ($p<1\times10^{-9}$); L2 (global integral) adds nothing over L1 (−0.1 N, n.s.); L3 (full anchor) adds 7.5 N over L2 ($p<1\times10^{-6}$). The L2→L3 step bundles four mechanisms at once, and the push metrics alone cannot separate them; Sec. 5.1.4 performs that separation on the idle side. What the push data do show is that the proximity gate contributes nothing to push *margin* on its own (L2→L2.5, +0.5 N, n.s.), consistent with its role being windup prevention rather than capture-basin enlargement, and that the full anchor recovers a margin advantage over the ungated integral (+7.5 N) while simultaneously being the only configuration in the ladder that settles. Subtractively, the striking result is how flat the ablation is: every single-mechanism removal except the pitch gate leaves $F_{\min}$ within 1.8 N of ACC, none of them significantly (0.087 – 0.48 uncorrected p). Capture basin is set by the anchor as a whole—the L2.5→L3 step—and not by any one of its four gates. This includes S0: gain scheduling is unnecessary, since fixing k_p=50 costs nothing measurable (−1.3 N, n.s.). It also includes the damping boost, whose value lies elsewhere. S2 is the dominant idle-precision mechanism and acts sagittally: removing it degrades sagittal repeatability sixteen-fold (0.294 mm→4.772 mm) while leaving lateral repeatability statistically unchanged (0.703 mm vs. ACC's 0.730 mm, Table 3), and its push margin is indis-

[1] Sweep: `scripts/sweep_push_chirality.py` (N=5 per bearing, four arms), raw output `outputs/push_chirality/`. The b_2 of that sweep's shipped arm (−16.2 N) and of row S1 of Table **??** (−16.3 N) agree, although the two use different seeds and settle protocols.

tinguishable from ACC's (80.8 N vs. 82.3 N, p=0.087). The boost therefore buys sagittal steadiness and settling, not capture basin. The one mechanism the push data *do* isolate is the pitch gate: removing it (S5) leaves the robot unable to survive the 10 N floor of the search in any of 80 trials, and additionally causes 10/10 falls at pure idle. S3 (asymmetric EMA) and S2 are each individually required for ringdown.

5.3. Flight Recovery and Terrain Adaptation

ACC's two-channel assembly supports independent extensions without modifying the core anchor. Both extensions use the same 400 Nm/s rate limit and 100 Hz control rate.

Contact-loss recovery. ACC recovers to a settled stand from a 100 cm free drop (10/10, 24.0±1.2° peak pitch after touchdown) and from a 50 cm ledge drive-off (10/10, 27.1±2.0° peak landing pitch); the 60 cm drop and 20 cm ledge are correspondingly milder (Table **??**). The two flight ablations, however, give a more nuanced picture than the mechanism's design rationale suggests. Removing the flight *attitude* PD (the reaction-flywheel term $2.0\theta + 0.5\dot{\theta}$, spin-down damping retained) is a real but modest loss: peak pitch rises from 24.0 to 27.8° on the 100 cm drop with one fall in ten, and from 27.1 to 31.6° on the 50 cm ledge with no falls. Removing the airborne wheel-torque override *entirely*—so the ordinary wheel–inverted-pendulum (WIP) loop keeps driving the wheels through flight—does not degrade either condition at all: it *lowers* peak pitch to 13.0±0.5° (drop) and 10.2±0.4° (ledge) at 10/10 survival, because free wheels driven by the pitch-feedback loop are themselves an effective reaction flywheel. The override's benefit appears only at the upper end of the envelope: at 150 cm the two configurations survive equally (2/5 settled either way) but the override holds peak pitch to 34.0±2.1° against 53.6±35.0° without it—a 16× reduction in spread, with one uncontrolled trial exceeding 90°. At 200 cm both configurations fail. The flight override therefore **bounds landing-attitude variance near the envelope limit** rather than enabling the recovery; at the 50 – 100 cm scales reported here ACC's balance core alone suffices.

Per-leg terrain adaptation. Here the ablation is unambiguous. With per-leg adaptation, straddle roll on the one-wheel curb is 2.5±0.04° (10 cm), 3.5±3.2° (15 cm) and 14.9±0.1° (20 cm), all 10/10. Replacing `split()` with a symmetric posture (both legs commanded to the same height—the pre-adapter behaviour) causes a fall at *every* tested curb height, 0/10 in all three cases: the torso rolls past 60° within 0.8 s of curb entry and the robot is on the ground by 2.3 s. The height split is thus load-bearing for this task in a way the flight override is not. The closed-loop leveling trim that rides on top of the split is a second-order refinement of the same mechanism: disabling it alone leaves the robot upright on the 10 cm curb (3.2±0.08°, 10/10) but degrades the 15 cm curb to 11.1±1.6° roll with 6/10 settled—the trim corrects the residual error of the non-linear height→leg-length table, which grows with step height. At 20 cm the limit is one of joint authority, not of the split geometry. Evaluating (21) at h^{cmd}=0.404 m gives commanded leg lengths of 0.350 m (flat) and 0.150 m (curb), i.e. a difference equal to the full 20 cm step with *zero* kinematic residual—the posture pair can span the curb exactly. What tightens is the joint budget: the compressed leg's commanded knee angle grows 2.18 rad (10 cm) → 2.39 rad (15 cm) → 2.58 rad (20 cm) against a hard limit of 2.70 rad, leaving only 0.12 rad (6.6°) of headroom, and the commanded CoM height of that leg, 0.265 m, sits within 1.1 cm of the extrapolation floor z_{lo}−10 cm=0.254 m. Because the commanded posture is exact, the measured 14.9° straddle roll is entirely *tracking* error: in this deeply flexed configuration the leg carries its load at the shortest gravitational moment arm the PID gains were never re-tuned for, and it settles below its target. Adaptation therefore still prevents leg-fold (10/10 survival), but the torso is left tilted; the tight ±0.1° std reflects a deterministic, saturation-dominated equilibrium rather than genuinely low run-to-run variability.

Oblique ledge descent: a straddle regime, not a height threshold. Driving off a ledge whose edge crosses the path at an angle ϕ releases the two wheels a stagger interval $w_{\text{track}} \tan\phi/v$ apart, and only within that interval can the leg split absorb the step. Instrumenting a 30°, 20 cm exit at 0.7 m/s shows the window closing in 0.18 s: the split had commanded 25 mm of differential leg travel against the 200 mm step when the trailing wheel lost contact, and the torso absorbed the remainder, reaching −33.1° of roll against the $\arctan(0.20/w_{\text{track}})$=35.7° that a wholly unspanned step predicts. The legs were not at their travel limit when contact was lost (99 mm of calibrated range unused), so the proximate failure is a race rather than a saturation—though relieving it would not rescue this case, because 20 cm is itself unspannable at nominal stance (170° of knee flexion against the 2.7 rad limit). The ±6 cm leveling trim is unavailable throughout, since its gate requires *both* wheels loaded. Failure is consequently *not* monotone in step height. Two regimes separate, set by whether the leading wheel reaches the lower ground before the trailing wheel releases—free fall over the step takes $\sqrt{2h/g}$, comparable to the stagger interval at these heights. A shallow step lands the leading wheel while the trailing wheel still bears on the platform, forcing the straddle the legs cannot span; a deep step releases both wheels together and is handled by the flight path of §4.5. Widening the straddle regime requires knee range rather than gains.

We calibrate the boundary at 192 trials—12 heights (12 – 40 cm) × 8 independent initial conditions × two angles, 43/96 and 24/96 settled (Table 5). Both regimes are real and the dead band between them is wide. At 30° the shallow window runs to 15 cm at 100 %, degrades through 17 cm (75 %) and 20 cm (25 %), and closes completely over 22 – 27 cm; the deep regime then reopens from 30 cm and peaks at 37 cm (75 %). At 45° the shallow window is essentially gone—12 % at 12 cm, zero from 15 cm to 25 cm—and the deep regime opens later, at 27 cm. That the window shrinks with angle is what the stagger interval $w_{\text{track}} \tan\phi/v$ predicts, and it shrinks by roughly the right factor. Two features are worth stating as measured rather than smoothed. Neither regime reaches 100 % at depth—the deep-regime maximum is 75 %, so a clean launch is a favorable initial condition and not a guarantee. And the 45° column has a reproducible 0/8 hole at 32 cm sitting between 50 % and 62 % neighbours, which the two-regime account does not predict. The boundary is therefore calibrated as a success-probability field over launch height rather than as a single threshold, and the 192-trial grid resolves that field directly.[2]

The gate structure isolates each extension to its physical regime without re-tuning the anchor, and both extensions cost three lines of controller code each. Their measured *necessity*, however, differs sharply: the per-leg height split is required for the curb task at every tested height, whereas the flight override is a variance-bounding refinement that only pays at the top of the drop envelope.

5.4. Comparison with Classical Baselines and a Preliminary RL Reference

To establish that ACC is not merely a tuned linear controller, we compare six classical baselines—four 4-state TWIP variants (LQR, LQR+AW, LQR-DT, PI+AW) and two 6-state coupled variants (servo and direct-torque)—together with a preliminary model-free PPO run reported as a reference point rather than as a seventh baseline (all share identical environment setups):

(1) **LQR**—4-state TWIP (Grasser et al., 2002; Pathak et al., 2005) with decoupled roll/yaw PD through a PID servo layer (~0.25 s lag).

(2) **LQR + Integral AW**—Same as (1) plus pitch-error integral with conditional AW (clamped ±3.0 rad/s).

(3) **LQR (Direct Torque)**—Re-optimized for direct-torque plant (τ_s=0, $k_{p,\text{roll}}$=55.0), outputting torques directly.

(4) **Coupled 6-State 3D LQR**—Extends (1) with roll states $(\phi, \dot{\phi})$.

(5) **PI + AW (Direct Torque)**—DT-LQR + integral AW on fwd. position (conditional, ±15 cm gate); same leg/roll/yaw as DT-LQR.

(6) **PPO reference (preliminary)**—Model-free policy trained from scratch for 5.4M environment steps in `BalanceEnv`: single seed, no domain randomization, no hyperparameter search. Included as a difficulty probe, not as a tuned RL baseline (Limitation 4).

(7) **Coupled 6-State LQR (Direct Torque)**—Same 6-state model as (4) but outputs direct torques (bypassing PID servo, ~0.25 s lag removed). Gains re-derived for the direct-torque plant (K_{pitch}=113, K_{roll}=58 Nm/rad); leg posture via PD position control.

Direct-torque LQR removes the servo path, eliminating both the PID gains and the measured actuation lag. PI+AW adds integral anti-windup on forward position to DT-LQR—the standard engineering fix for a DC position bias, with windup protection. The anti-windup logic is verified independently by `scripts/validate_lqr_aw.py`. The PPO run serves a different purpose from the six classical controllers. It is included not to benchmark the state of the art in reinforcement learning, but to characterize how hard unaided model-free search is on the coupled pitch–height state space of a 10-DOF wheeled biped, and

[2] `scripts/ramp_step_tests.py` with `--seeds`; raw output `outputs/oblique_regime/diag30_fine.txt` and `diag45_fine.txt`. Each seed perturbs the initial joint angles by 3 mrad and the root by 1.5 mm, so the eight trials per cell sample initial conditions on an otherwise deterministic course.

thereby to make explicit the value of the structured gate design ACC supplies in its place. Every gain used for the six classical baselines is listed in Table 7; we give them in full because the central claim of this subsection is a *negative* one—that no member of this family stands—and a negative result is only as strong as the tuning behind it.

Table 8 reports the comparison. All six classical baselines—four derived from the 4-state TWIP model (LQR, LQR+AW, LQR-DT, PI+AW) and two 6-state coupled variants (servo, direct-torque)—fall within 0.60 s. Removing the PID servo path (4-state DT-LQR: τ_s=0, $k_{p,\mathrm{roll}}$=55.0, no integral) cuts torque by 2.5× (21.99 Nm → 8.76 Nm) and nearly doubles survival (0.32 s → 0.60 s), but *worsens* roll (ϕ_{RMS}=27.5° vs. 24.2°): the recovered authority is spent in the sagittal plane the model describes, not in the lateral one that ends the episode. Adding a forward-position integral (PI+AW: $k_{i,\mathrm{pos}}$=8.0, conditional ±15 cm gate) gives back 0.08 s of that gain rather than adding to it—the integral cannot accumulate usefully before the roll-induced fall—confirming the 4-state TWIP model is structurally insufficient for this articulated-leg platform. The preliminary PPO reference (5.4M steps, single seed) sustains single-height standing longer (3.20 s) but exhibits an 85 % fall rate across multi-height command transitions (54.2 mm height error). Read as a difficulty probe rather than a comparison, this is the observation the structured prior is meant to address: unaided search spends its budget rediscovering the pitch–height coupling that the posture map and the gate schedule encode directly. ACC achieves 0.060±0.006° pitch RMS, 0.010±0.001° roll RMS, 0.18±0.02 mm CoM Z, and 3.28±0.00 Nm torque RMS (N=10). A **46-configuration** sweep (dimensions as given in Table 7; script `scripts/sweep_lqr_roll_gains.py`) confirmed all tested LQR variants fail (100 %); a 6-state coupled LQR with direct torque actuation, removing the PID servo path, also fails—indicating platform complexity, not servo lag alone, is the dominant factor (see Limitation 4). Neither servo-lag removal nor preliminary PPO training (5.4M steps, single seed) resolved the height-adaptive posture trade-off in our experiments; whether a fully converged PPO policy closes this gap is not tested here.

5.4.1. Both control rates give the same outcome

Table 8 reports every classical baseline at ACC's 100 Hz, so the headline comparison carries no control-bandwidth differential. Those gains were tuned for a 50 Hz loop, however, and a reader is entitled to ask whether running them off their design rate is what produces the gap. Table **??** answers that by giving both rates side by side.

Doubling the baselines' control rate does not rescue any of them. Fall rate remains 100 % in every cell, and survival time does not improve for a single variant—it *shortens* for five of six (LQR: 0.52 s→0.32 s; LQR-DT: 0.72 s→0.60 s; PI+AW: 0.72 s→0.52 s) and is unchanged for the sixth. The diagnostic quantity is roll: ϕ_{RMS} is 24.21° at both rates for the three servo variants, and rises from 21.22° to 27.48° for LQR-DT. Faster sampling improves the quantities the controller actually regulates—pitch RMS falls by roughly 6× for the servo variants (0.037°→0.006°) and τ_{RMS} drops from 92.1 Nm to 22.0 Nm—while leaving the lateral divergence that terminates the episode untouched. This is the signature of a model-structure deficit rather than a bandwidth deficit: the 4-state TWIP model carries no lateral state at all, and the 6-state coupled variant carries roll but linearizes it about a single operating point, so no sampling rate supplies the missing dynamics. Reporting the baselines at 100 Hz in Table 8 is the rate-matched choice, and it is deliberately not the flattering one for them: five of the six survive *longer* at 50 Hz. Both rates are given here so that the comparison can be read either way, and the fall rate is 100 % in all twelve cells.

The converse experiment—running ACC at 50 Hz—is a negative result: the controller does not survive the rate change, and it is recorded as a limitation (§6, Limitation 2) rather than as a rate-independence claim. What Table **??** establishes is the direction that matters for the comparison: the classical baselines fail both at the rate for which they were designed and at ACC's rate.

5.4.2. A full-state LQR designed on the plant itself

All six baselines above are designed on a reduced model, so their failure is open to the obvious objection that the model, not the method, is what fails. The control experiment that settles this is a linear-quadratic regulator derived from the plant directly: a damped Gauss–Newton solve for a true static equilibrium $(\mathbf{x}^*, \mathbf{u}^*)$ at the commanded height, central differences on the whole held-input control step of the complete 16-DOF, 10-actuator MuJoCo model for the transition Jacobians, removal of the two cyclic wheel-angle states, and a discrete-time Riccati solve on the remaining 30. Torque is emitted under the same 400 Nm/s rate limit as every other row, and Bryson's rule (Bryson & Ho, 1975) fixes **Q** and **R** from physical maximum deviations, leaving one knob—a uniform control weight r—swept over six decades rather than tuned. The design is sound on every diagnostic that bears on it: at the 0.65 m equilibrium the residual unbalanced force is 0.428 N against 79.46 N of body weight (0.54 %), the reduced **A** has full rank 30 and is PBH-stabilizable, the Riccati relative residual is $\sim 10^{-14}$, and the designed closed loop has max $|\lambda|$=0.9911. The one figure that invites a disqualifying reading, $\kappa(\mathbf{A})\approx 10^{10}$, does not support it: it is a 50 Hz artifact (2.6×10^{7} at 100 Hz) traceable to a $\sim 10^{-9}$ *smallest* singular value—strongly *contracting* contact modes—and it obstructs neither the Riccati solve nor the resulting feedback. Whatever limits this controller, it is not the linearization.

What the sweep shows instead is a slow divergence that a short horizon hides. Over 20 s the design looks successful: at h=0.65 m, r=1000 gives 0 % falls with 11.0 mm height RMSE and 0.12° pitch RMS. At 60 s every configuration tested falls in 100 % of episodes, at 22.2 – 57.0 s, with planar drift growing monotonically to 0.43 – 0.63 m—that drift, not attitude, is what ends them. Nor does one weight cover the band: r=100 stands 20 s at 0.60 m and 0.69 m but falls at 0.65 m, r=1000 does the reverse, and under uniformly random height commands in 0.40 – 0.70 m the fall rate runs 45 – 80 % across the three usable weights. At the baselines' own 50 Hz rate every weight falls inside 20 s. Push separates it from ACC outright: 100 % falls under the standard 50 N impulse, and a maximum recoverable push of 10.0 – 20.2 N against ACC's $F_{\min}$=82 N. The reading is therefore not that the plant defeats linearization—it does not—but that full-state optimal linear feedback about a single equilibrium buys roughly two orders of magnitude of survival over the reduced-order designs (0.16 – 0.60 s → 22 – 57 s) and still does not close the problem: it diverges slowly rather than quickly, holds one height at a time rather than the band, and has essentially no push envelope. Two of those three are exactly what the anchor integral and the gate schedule are for, which is why the full-state design is reported as a control experiment rather than promoted into Table 8—at the 20 s horizon used for every other row it would read as a partial success, and that reading is an artifact of the horizon.[3]

5.4.3. *Sensitivity of the coupled model's empirical roll coupling*

The two 6-state coupled baselines close their roll channel with a constant $b_{r,h}$ (hip-roll angle to roll acceleration) that the planar pendulum model does not supply in closed form; it was set empirically to −5.0. Since the negative result of this subsection rests on those rows, that constant is swept rather than assumed: nine values from −0.5 to −40 (an 80× range spanning both sides of the nominal), gains re-derived through the same Riccati solve at each value, both variants, N=20 episodes × seeds $\{0, 42, 123\}$ at 100 Hz.

The outcome does not move. The servo variant falls in 100 % of episodes at 0.320 s for every value, with roll RMS between 24.209° and 24.224° (a spread of 0.06 %); the direct-torque variant falls at 0.160 s for every value, with roll RMS 21.186 – 21.187°. This is invariance to the parameter, not insensitivity of the metric: the re-derived roll gain varies 32× across the sweep for the servo path (2.24 – 72.71) and 8× for the torque path. The mechanism is saturation. The roll-channel command reaches its ceiling—the ±0.7 rad hip-roll joint limit for the servo path—at a roll excursion of 0.55 – 17.87° depending on $b_{r,h}$, while the roll excursion actually observed before the fall has an RMS of 24.2°. The channel is therefore commanded against its limit throughout the episode at every value in the sweep, and the gain that computed the command cannot matter. Raw output: `outputs/b_roll_hip _sweep/results_100hz.json` (`scripts/sweep_b _roll_hip.py`).

5.5. *Whole-Body Control Baselines*

To address whether a tuned whole-body controller (WBC) could match ACC, we evaluated two WBC configurations that were developed and subsequently discarded during controller development: **(W1)** a task-priority quadratic-program (QP) WBC—minimizing COM height error, torso orientation error, posture damping, and wheel regularization subject to full-body dynamics and contact constraints—used as the *primary* balance controller; and **(W2)** the same WBC

[3] N=20 episodes × seeds $\{0, 42, 123\}$ per configuration at 100 Hz; between-episode SD is 0.00 s on time-to-fall, and weights r<100 fall within 1.4 s. Produced by `scripts/eval_balance.py--controllerbaseline_full_lqr`; per-configuration results and the design diagnostics are in `outputs/fullstate_lqr_corrected/`, whose `README.md` maps each row to its run.

blended as an *assist* on top of ACC via $\boldsymbol{\tau}_{\text{assist}} = \boldsymbol{\tau}_{\text{ACC}} + \alpha \cdot (\boldsymbol{\tau}_{\text{wbc}} - \boldsymbol{\tau}_{\text{ACC}})$, evaluated both with the per-joint authority gate active and with it disabled. Both were evaluated under identical cloned simulation conditions (three-arm counterfactual: ACC baseline, WBC-only, ACC+WBC assist; 225 scenarios spanning fixed-height balance, height transitions, and push recovery at 20 – 120 N; 481 000 control steps; ACC profile `K2_JAX_DEDICATED_DEFAULT_V3`; no gain retuning). So that the comparison is not a verdict on our own implementation, the WBC was verified against executable references before evaluation—its analytic contact Jacobian against MuJoCo's `mj_jac`, its task stack against each of four independently selectable weightings, and its height reference against the CoM-z convention of §3.1 (Appendix B, Table 17). Every number below comes from that verified implementation.

W1 (WBC as primary controller). The standalone QP-WBC falls in **225 of 225** scenarios; ACC falls in **none**. It holds the robot upright for 0.570 s (SD 0.001 s) and spends 97.3 % of every episode in a fallen state (468 179 of 481 000 control steps). The failure is *lateral*: roll RMS is 93.45° against ACC's 3.71° (25.2×) and peaks at 103.9°—past horizontal—while pitch RMS is only 2.33° against 0.52° (4.5×). Base height collapses from 0.540 m to 0.091 m. The fallen fraction is uniform across all five scenario suites (97.2 – 97.4 %), so the collapse is not specific to pushes or to commanded height changes.

The failure is invariant to implementation quality. Solver quality is not what decides the outcome, and a control experiment isolates it: a degraded build of the same controller, whose QP returns a usable solution on only 85.85 % of steps, falls at 0.52 s, while the verified build—99.99 % solved (480 950 of 481 000 steps), commanded torque RMS 0.96 Nm and max 4.33 Nm—falls at 0.570 s. A 14-point improvement in solve rate buys 50 ms of upright time against a 0.570 s lifetime. Nor is the failure an artifact of one task weighting: re-running the WBC-only arm at four task-weight modes over an identical scenario set (Table 15) moves the fallen fraction by 0.1 percentage points and upright time by 0.03 s. The best mode, `torso_priority`, improves pitch RMS by 40 % but *worsens* roll—within a single SO(3) orientation task carrying equal roll and pitch gains, the QP trades one axis against the other rather than fixing both.

Mechanism. The WBC's posture task is a velocity damper, not a position tracker. Its reference $\mathbf{q}_{\text{ref}}$ is rebuilt from the currently measured joint vector at every control step, so $\mathbf{q} - \mathbf{q}_{\text{ref}} \equiv \mathbf{0}$ identically and the task collapses to $-k_{d,\text{posture}}\dot{\mathbf{q}}$ with $k_{d,\text{posture}}$=2.0. It can slow a postural drift but cannot reverse one. Combined with the absence of gravity-compensation feedforward, nothing in the task stack restores the legs once a lateral excursion begins—which is why the collapse is lateral, why it is insensitive to the task weighting, and why solver quality does not delay it. Critically, this is not a matter of tuning or of implementation: the WBC lacks the *regime-gating* that ACC's proximity-gated anchor provides. A WBC distributes torque optimally for a fixed task set—it cannot express qualitatively distinct physical regimes (quiet stance, push ringdown, large excursion) as distinct control modes. The precision–recovery trade-off ACC resolves through spatial gating and asymmetric temporal envelope-following is not expressible as a task-priority QP objective; even an ideally implemented WBC would lack the structured prior that gates the anchor integral, and would therefore face the same global-integral windup that our ablation S4 (no proximity gate, Table **??**) confirms is fatal for sub-millimeter precision.

W2 (WBC as assist on ACC). We ran the assist in both of its available configurations, which together constitute a gate ablation (Table 10).

Gate disabled (α=0.25 fixed, per-joint authority capped at 20 % of the ACC torque; this is the configuration behind the 225-scenario campaign). The QP solved on 100.0000 % of steps with zero fail-closed events, so WBC torque entered the blend unconditionally. The assist arm never falls—ACC carries it—but it degrades ACC on every metric except one: pitch RMS 1.295° vs. 0.520° (2.5×), yaw drift 12.56° vs. 6.69° (1.9×), planar drift 0.199 m vs. 0.099 m (2.0×), height RMS 0.518 m vs. 0.540 m, and torque RMS 3.67 Nm vs. 3.18 Nm.

Gate enabled (per-joint authority $\alpha_j = \alpha_{\max} \cdot g_{\text{stability}} \cdot g_{\text{height}} \cdot g_{\text{push}} \cdot g_{\text{divergence}} \cdot A_j \cdot K_{\text{role},j}$, $\alpha_{\max}$=0.25). We instrumented the gate to record α and every factor at every control step. On the one height variant where the gate opens at all, the admitted authority is $\bar{\alpha}$=0.0061 (max 0.0146): 2.5 % of the permitted $\alpha_{\max}$, peaking at 5.8 %. On the remaining four variants the height factor g_{height} falls below its 1×10^{-3} cutoff on the commanded height alone and α is identically zero (Appendix B, Table 16). The gated assist is consequently indistinguishable from ACC alone on identical scenarios—pitch RMS 0.519°

vs. 0.520°, torque RMS 3.154 Nm vs. 3.154 Nm, height RMS 0.5412 m vs. 0.5412 m—agreeing to four significant figures because the gate admits almost nothing. The two configurations bracket the assist: where WBC authority is admitted it degrades ACC, and where it is harmless it is inactive. No setting of the gate makes the assist simultaneously active and beneficial.

The one metric WBC improves. Roll is the exception in both configurations: with the gate disabled, roll RMS improves from ACC's 3.71° to 3.20° over the full batch (−14 %), and from 3.55° to 2.90° on the matched subset (−18 %). The improvement is systematic rather than sampling noise: it appears in both the full batch and the matched subset. ACC regulates the lateral axis with decentralized hip-roll PD, whereas the WBC uses a centralized orientation task that exploits full mass-matrix coupling; on that axis alone, the centralized formulation is the better regulator, and blending the two writes both into the hip-roll joints from incompatible premises. What it cannot do is stand: its posture stack has no position authority, so buying 18 % of roll costs 2.5× the pitch error and 2.0× the drift. A centralized lateral task is therefore a credible direction for improving ACC's weakest axis—the 98.2 N mean lateral push bearing of Sec. 5.2—but not through torque-level blending with a posture stack that cannot hold a posture.

Timing. The QP solve costs 0.169 ms mean and 0.30 ms p99 over 481 000 solves (OSQP (Stellato et al., 2020), offline Python/NumPy path), comfortably inside the 10 ms control period. Compute is therefore not the constraint. The offline matrix-assembly path surrounding the solver is un-optimized Python, and production C++ whole-body controllers run at 400 – 1000 Hz on embedded hardware. The assist is excluded on control grounds alone: it degrades every stability metric but one wherever it is active, and does nothing wherever it is not.

Summary. Neither standalone WBC nor WBC-as-assist resolves the precision–recovery trade-off on this platform. The trade-off is resolved by regime-gating—the proximity-gated anchor and asymmetric envelope follower—which WBC's task-priority formulation does not express. ACC's conditional gates therefore provide the essential structured prior; the WBC gap is one of structural expressiveness, not torque optimality.

5.6. Architectural Analysis

We evaluate ACC under four dimensions: noise/delay robustness (Table 11), gate parameter sensitivity, torque rate-limit invariance, and compound-disturbance recovery.

Robustness to sensor noise and actuator delay. A factorial sweep over four noise levels and three delay levels (0 ms, 10 ms, 30 ms; N=20 per cell, 240 trials) bounds ACC on both axes at once. One control period of delay is benign for stability and expensive for performance: at 10 ms there are zero falls on clean and low noise, but $F_{\max}$ bifurcates from 95.7 N to 50.5 N—roughly half—and idle RMS rises 5.6×, from 0.43 mm to 2.43 mm. Three control periods is a different regime entirely: at 30 ms the platform falls in 20 of 20 trials on *clean* sensing, and equally under low and medium noise, with no recoverable push at any tested magnitude. Delay tolerance is therefore bounded rather than open-ended, and the bound lies between one and three control periods; §5.6 locates it to 14 – 16 ms. Noise alone is survivable up to the medium level (0 falls in 20 at 0 ms delay). What ACC does not tolerate is the *conjunction*: medium noise together with 10 ms of delay costs 2 falls in 20 trials, the only sub-catastrophic falls anywhere in the grid—every other fall in the table is a cell in which all twenty trials fail. High noise is unrecoverable at every delay (60/60, including at zero delay); the asymmetric envelope cannot distinguish noise from motion, forcing anchor disengagement, and the residual P-only mode cannot hold the platform. That conjunction is the binding constraint on the design, and §C shows it is what prices the one gain change that would otherwise remove ACC's parking offset. Idle repeatability degrades from 0.43 mm (clean) to 30.82 mm under medium sensor noise at zero delay—a 71-fold loss, and the sharpest single sensitivity in the design. The mechanism is the one described above: the envelope gate cannot separate sensor noise from real motion, so it releases the integral and re-engages it repeatedly. This locates the boundary of the anchored regime precisely. Below it the anchor delivers the precision of Table 3; at medium noise it no longer pays for itself, the un-anchored L0 baseline holding R_{lat} = 3.675 mm on the same axis. The design requirement that follows is explicit and testable: ACC is specified against a filtered velocity estimate, which is a requirement on the state estimator rather than a limit on the controller. Measurement conventions for this table: the idle column is *lateral* (world x), not the sagittal axis of Table 3, and uses a 3 s rather than 5 s settling allowance, so its clean reference is the 0.43 mm entry and not ACC's lateral $R_{\text{lat}} = 0.730$ mm (with EMA_0=0 an earlier window captures more of the strongly-anchored transient); it averages surviving trials only, so the falls column is read first; and every cell is measured identically, so the degradation *factors* are the quantity this table establishes.

Resolving the delay axis. The grid above quantises delay to the control period, so it brackets the useful range at 10 ms and 30 ms but says nothing about where inside that bracket the behaviour changes; the knee was therefore located separately with the transport delay applied per *physics* substep (2 ms at 500 Hz). This resolves the axis five times more finely and is also the more faithful model, since a real actuator's latency is not quantised to the controller's period. On clean sensing the harness is deterministic—every multi-seed cell returned identical values across all seeds—so single-seed cells are exact rather than sampled. Table 12 places the knee between 6 ms and 8 ms: $F_{\max}$ is flat to within 2.4 % out to 6 ms and then collapses by 45 %. Beyond the knee the decline continues monotonically—41.6 N at 12 ms, 28.8 N at 16 ms, and the search floor by 20 ms, i.e. no recoverable push at all. The tail matters because it distinguishes a genuine loss of margin from a narrow phase-alignment notch, and it is the former. Two consequences follow. First, the transition is a cliff rather than a roll-off, so a design either clears it or does not; there is no graceful degradation to trade against. Second, the knee sits *below* the 10 ms figure the coarse grid implied, and therefore below the 9.5 ms end-to-end budget estimated in §6—the sign of that margin is negative, not positive (Limitation 2). Absolute levels are not directly comparable with Table 11, which uses a different push protocol, but the two harnesses agree on both anchors: 95.5 N vs. 95.7 N at 0 ms, and 56.9 N vs. 50.5 N at 10 ms—in both cases a post-cliff value near half the nominal one. What this sweep establishes is the location of the knee, which the coarse grid cannot see.

Losing the push margin and losing the platform are different thresholds. The cliff above is a loss of *margin*: past 8 ms the robot still stands, it simply stops absorbing pushes. The delay at which it stops standing at all is a separate and higher quantity, and we measure it with the same substep operator applied to a quiet-stance trial (3 s settle + 20 s record, clean sensing, `scripts/delay_retune_sweep.py--idle`). ACC holds station through 14 ms, though visibly degraded—idle lateral RMS runs 2.47 mm at 10 ms and 10.84 mm at 14 ms, against 0.43 mm undelayed—and falls at 16 ms and beyond. The threshold is thus 14 – 16 ms, bracketed to one physics substep, which is the finest this harness resolves. Above it the collapse is prompt (fall at 1.8 s at 18 ms, 0.7 s at 30 ms), which is what makes the 30 ms column of Table 11 unanimous. One consequence bears on how the 16 ms entry of Table 12 should be read. At that delay the quiet-stance trial diverges only at 22.0 s, later than the push harness's 20.1 s horizon, so the 28.8 N recorded there is measured on a platform that is already unstable and merely slow about it. The last delay at which $F_{\max}$ describes a platform that also survives the full quiet-stance window is 14 ms; the practical envelope is set by the smaller of the two thresholds—the 6 – 8 ms cliff—in any case.

Moving the knee: a delay-hardened variant. Whether the knee is a property of the architecture or of the shipped gain set is a question the sweep can answer directly. Holding the delay at 8 ms—just past the knee—and sweeping one profile constant at a time gives Table **??**. The result is not a gently sloped optimum but a sharp *local dip*: the shipped value of every one of the four constants tested scores 51.0 N, while nearly every neighbouring value, in *either* direction and on *any* of the four, recovers 78 – 93 N. A margin that is genuinely exhausted does not behave that way. The shape is the signature of an unlucky phase alignment at one operating point, and it means the 6 – 8 ms knee is a tuning artefact rather than an architectural ceiling.

Carrying the largest single-knob change forward—sagittal velocity damping doubled from 22.5 to 45 Nm/(m/s), denoted ACC-DH—gives the second row of Table 12. ACC-DH is identical to the shipped controller at 0 – 2 ms, no worse at any delay tested, and moves the knee outward by one grid step, from 6 – 8 ms to 8 – 10 ms: it holds 93.2 N at 8 ms where the shipped gains give 51.0 N, and adds 44 – 90 % of margin across 12 – 16 ms. Both arms reach the search floor by 20 ms, so the retune buys headroom below the cliff and does not relocate the eventual failure. The residual dip at 10 ms (60.4 N) is the one delay in the sweep that is a *pure* one-control-period hold; every other cell mixes two consecutive commands within a control period, so it is not directly comparable with its neighbours.

The measured price is small at zero delay and real under delay. Idle lateral RMS undelayed rises from 0.790 mm to 0.801 mm (+1.4 %, N=3), inside the run-to-run spread of the quantity; but with delay applied ACC-DH is the less steady of the two, 2.89 mm vs. 2.47 mm at 10 ms and 14.59 mm vs. 10.84 mm at 14 ms, and it falls at the same 16 ms as the shipped gains. The retune therefore moves the push cliff without moving the standing threshold—the two are decoupled, and the 16 ms figures quoted above for both arms carry the horizon caveat of the preceding paragraph. Under the noise-plus-delay conjunction that produces the only sub-catastrophic falls in Table 11, ACC-DH is *better* rather than worse: at 8 ms with N=5, $F_{\max}$ rises from 76.8 N to 92.0 N under low noise and from 84.3 N to 90.2 N under medium, and the seed-to-seed spread contracts sharply (16.2 N to 1.2 N standard deviation under low noise). The same table exposes a separate point worth stating: the shipped controller's 8 ms dip is far deeper on clean sensing (51.0 N) than under noise (76.8 N and 84.3 N), so sensor dither de-phases the deterministic worst case. The clean-sensing cliff of Table 12 is therefore a conservative bound, not an optimistic one.

We report ACC-DH as an evaluation variant and do *not* adopt it. The constant it changes is the ungated damping that also governs driving acceleration and settling, and its cost on the driving, terrain and flight axes has not been re-measured; the 48-scenario qualification suite, not a single-axis sweep, is what decides a default. The claim made here is narrower and sufficient for the limitation it addresses: the delay knee is movable by tuning, at negligible cost to undelayed quiet stance and with a measured *gain* under noise—but it buys push margin only, and leaves the 14 – 16 ms standing threshold where it was.

Gate parameter sensitivity and tuning. Finite-difference gradients reveal a stark hierarchy: the **release coefficient** α_r **dominates** (sensitivity 220.4, 20× the next, 130× pitch gates); pitch stability thresholds (≈0) are effectively insensitive safety interlocks. For porting to a new morphology: (1) identify dominant ringdown period T_r from post-push decay; (2) set $\tau_{\text{release}}\approx 3{-}5\times T_r$; (3) set quiet-stance band 2–3× idle velocity noise floor; (4) set pitch gates conservatively (±12°, ±15 °/s); (5) tune k_p (fixed k_p=50 sufficed). A fixed k_p and anchor integral gain K_i require minimal adjustment across platforms; the critical parameter is τ_{release}.

Gate failure modes. Stuck-at-1 ($g_{\text{anchor}}\equiv 1$) causes windup → fall (cf. S4); stuck-at-0 yields P-only limit cycle (cf. L0); stuck-half-open triggers parametric resonance (cf. S5, no survivable push at the 10 N search floor). All three map to measured ablation rows (Table **??**), providing implicit FMEA coverage.

Torque rate-limit invariance. The 400 Nm/s slew limit used throughout is not what sets ACC's push ceiling. Sweeping it over an 8× span, with each trial drawing an independent initial-posture perturbation (the same $\mathcal{N}(0, 0.005\,\text{rad})$ joint / $\mathcal{N}(0, 0.001\,\text{m})$ root draw as the push protocol of §5.2, N=5 per setting), gives lateral $F_{\max}$ of 92.5±1.1 N (100 Nm/s) and 94.4±0.0 N at 200, 400 and 800 Nm/s, with idle lateral RMS of 0.695±0.046, 0.738±0.031, 0.732±0.025 and 0.729±0.023 mm respectively—a 2 % spread in push margin and a statistically indistinguishable spread in idle precision. The limiter is genuinely active where it could matter: instrumenting the commanded torque during a 90 N push shows $|\dot{\tau}|$ saturating exactly at each configured bound (100, 200, 400 Nm/s; at 800 Nm/s the wheel channel still reaches the bound while the leg channels peak at 712 Nm/s). During quiet stance, by contrast, the peak commanded slew measured across the four settings is 0.51 – 1.66 Nm/s—60 – 190 times below even the 100 Nm/s bound—so idle precision cannot be slew-limited at any tested setting. The ceiling is therefore architectural rather than actuator-limited.[4]

Compound-disturbance recovery. The push protocol of §5.2 holds the height command fixed. Because the posture map is height-scheduled, a commanded transition and a push compete for the same leg-torque budget, so the two disturbances are evaluated together here. Scenario A applies the standard lateral push impulse at the midpoint of a 1 s commanded height ramp of size Δh about the 0.404 m nominal CoM-z, and locates $F_{\max}$ by the same binary search; Δh=0 is the matched static control, run under the identical protocol so that the two differ only in the height command. Every trial draws an independent initial posture from the §5.2 distribution, N=10 throughout. A commanded height change must move *both* the gain schedule and the leg posture target; ramping the schedule alone would leave the robot standing at its nominal height with detuned gains, which is a different experiment. Both are ramped here, and the achieved CoM height is verified against

[4]Sweep: `scripts/sweep_rate_limit_corrected.py`, raw output `outputs/rate_limit_sweep/results_corrected.json`; slew instrumentation: `scripts/probe_torque_slew.py`.

the model's own forward kinematics before each sweep. The ramp is applied as a delta about each trial's own settled posture, which makes the Δh=0 row bit-identical to the static control. Table 14 reports the result.

Three findings follow. First, *inside the calibrated posture band a simultaneous height command is nearly free*: a +5 cm rise costs 5.1 % of push margin (89.9 N vs. 94.8 N) and a −5 cm squat is *better* than static (101.0 N, +6.5 %), because squatting lowers the CoM and shortens the toppling moment arm while the push arrives. Both effects are statistically resolvable but operationally small, and they bracket the static value—there is no shared-budget penalty at this amplitude. Second, the degradation is one-sided, and *descent is free at any amplitude tested*. Squatting −10 cm—fully outside the calibrated band—costs nothing measurable (94.8 N vs. 94.8 N, p=0.99), while rising +10 cm costs 68.7 %. Extrapolation *per se* is therefore not the mechanism; direction is. Rising drives the knee toward extension, where the leg Jacobian is worst conditioned and the same joint torque produces the least vertical force, and it does so while raising the CoM; squatting moves in the opposite direction on both counts, and the −10 cm row pays for it only in peak pitch (28.9°, the largest in the table) rather than in margin. A control experiment separates how much of the remaining collapse is calibration error and how much is the physics of standing tall. The shipped posture map is exact inside its band but drifts outside it, undershooting the commanded CoM height by 10.0 mm at 0.504 m and 29.1 mm at 0.554 m. Re-parameterizing the *same* posture family so that the achieved height matches the command—no controller change, no new configurations, only a relabelling—raises $F_{\max}$ at +10 cm by 78 %, 29.7 N→52.8 N ($p=1.5\times10^{-4}$), and drops the peak pitch from 12.6°±13.1 to 8.8°±2.3. It does not restore the static value: 52.8 N is still 44 % below it, so the relabelling closes 35 % of the 65.1 N deficit. The calibration error is thus a real and removable contributor, but a minority one; the residue is the genuine cost of carrying the load on a more extended leg. Inside the band the two maps are indistinguishable, which is the control the arm needs to be interpretable. The large peak-pitch spread of the shipped +10 cm row (±13.1°, one trial reaching 46.7°) is consistent with trials sitting on the edge of the capture basin rather than comfortably inside it, and it is that spread—not the mean—that the re-calibration removes. Third, Scenario B tests sequential rather than simultaneous disturbance: a 90 N lateral impulse followed 2 s later by a 60 N counter-directed impulse, before the first ringdown has fully decayed. ACC survives **10/10** (Clopper–Pearson 95%CI [0.69, 1.00]) with an episode peak pitch of 11.89 ± 0.51°, essentially all of it from the first impulse: after the reversal the pitch peaks at only 2.78 ± 0.11° and never re-enters the 5° band used as the settle criterion, in every trial. The reversal is thus absorbed within the residual motion of the first recovery, which is the behavior the asymmetric envelope is designed to produce—$\tau_{\text{release}}\approx1.5$ s keeps the damping boost still partly engaged when the second impulse arrives.

6. Discussion and Conclusion

ACC achieves all demonstrated behaviors without neural network training, is hypothesized to provide a *reduced* sim-to-real transfer gap vs. learned policies—pending hardware validation (Grimminger et al., 2020). Each gate was added after tracing a measured failure mode to its root cause: (i) bistable envelope; (ii) global-I windup; (iii) parametric pumping; (iv) hard-leash oscillation; (v) self-referential stiffness. All thresholds frozen prior to the ablation sweep (Table **??**).

Statistical power. Welch's t-test, Bonferroni-adj. α=0.007 (7 comparisons); L0→L1 and L2.5→L3 significant (p<0.001; Cohen's d=5.4 and 4.1). The primary idle claims are stated as like-for-like ratios in Sec. 5.1 (59.4× sagittal repeatability, 22.6× planar repeatability, 5.0× lateral repeatability, all L0 vs. ACC at N=10 under one protocol; the headline is the sagittal figure, which is the actuated axis); the S5 collapse is unambiguous (0/10 survival at idle, 10/10 at every push magnitude). Flight/terrain: N=10, Clopper–Pearson 95%CI [0.69, 1.00].

Limitations. (1) **Simulation only.** All results are MuJoCo; hardware validation remains future work, and the sim-to-real gap is hypothesized to be reduced relative to learned policies rather than measured. (2) **Deployment timing is the binding constraint.** ACC as tuned requires a 100 Hz loop and an actuator latency clearing the 6 – 8 ms push-margin cliff of Table 12; it stops standing by 14 – 16 ms, and the 9.5 ms end-to-end budget estimated below sits past the cliff. Rate-independence is not claimed: at 50 Hz with coefficients recomputed for the longer step ($\alpha_a\approx0.58$, $\alpha_r\approx0.013$) the platform does not stand (mean CoM height 0.124 m against 0.404 m nominal). Whether that is excitation of plant modes the faster loop suppresses or phase lag from discretizing the asymmetric envelope at Δt=20 ms is not resolved here; it requires a loop-gain and phase-margin analysis. Both thresholds are measured on clean sensing, with the noise×delay conjunction bounded only by the 10 ms and 30 ms columns of Table 11. (3) **Two calibrated envelopes are non-uniform.** Oblique ledge descent has two survivable regimes separated by a dead band (Table 5, 192 trials); the best cell is 75 %, so a clean launch is a favourable initial condition rather than a guarantee, and the 45°/32 cm cell is a reproducible 0/8 the two-regime account does not predict. Upward height commands under simultaneous push are restricted: a +10 cm rise costs 69 % of $F_{\max}$, while descent is free at every amplitude tested. A control experiment attributes 35 % of that deficit to posture-map parameterization and the remainder to the physical cost of standing on a more extended leg; the re-parameterization is not shipped, since changing the posture map would require re-qualifying the standing suite against which it was frozen. (4) **The learning comparison is not resourced, and one classical family is untested.** The model-free PPO run (5.4M steps, one seed, no domain randomization (Tobin et al., 2017; Peng et al., 2018) or hyperparameter search) is a difficulty probe; its 85 % fall rate is reported as observed and bounds nothing about RL on this platform. The comparison this controller is built for is residual learning over the prior, not against it (§1). Predictive control is the remaining untested classical family: a tuned convex MPC over the linearization of §5.4.2 is the natural next comparator, and the slow drift divergence that limits the full-state LQR is what a receding-horizon formulation with a terminal cost addresses. The task-priority QP whole-body controller does not resolve the precision–recovery trade-off (§5.5); roll RMS is the single axis on which the blend beats ACC (14 %), so the centralized formulation has merit that torque-level blending cannot harvest. Push ablations at N=10 resolve ~2 N, set by between-episode dispersion rather than by the search bisection; no claim in this paper rests on a smaller difference. (5) **Generality across morphologies is undemonstrated.** Every result is for one 10-DOF platform with one set of gains. The porting recipe of §5.6 is a hypothesis derived from this platform's sensitivity structure, not a validated procedure.

Deployment Risks and Mitigations. ACC uses raw torque, 400 Nm/s rate limit, standard sensors (IMU + encoders). The binding latency figure is a **performance cliff between 6 ms and 8 ms**, across which $F_{\max}$ falls 95.5 N→51 N (Table 12); below 6 ms the cost is under 2.4 %. The companion stability threshold is roughly twice as far out: on clean sensing ACC holds station through 14 ms and falls by 16 ms (§5.6). That ordering is the one the architecture predicts—ACC's gates depend on physical state rather than timing precision, the proximity gate being spatial (15 cm), the asymmetric envelope temporal but slow (τ_r≈1.5 s), and the damping boost state-driven rather than schedule-driven—but "twice" is a factor of two, not an order of magnitude, so a build that misses the latency target does not merely lose push margin gracefully. Commission against the 6 ms cliff, not against the 14 ms fall threshold.

Estimated non-RTOS worst-case budget: sensor readout ~1 ms + state estimation ~2 ms + ACC computation ~3 ms + actuation latency ~3.5 ms = ≈9.5 ms end-to-end—a budget assembled from assumed component figures, not a measurement. Read against a cliff at 6 – 8 ms, that budget is *over* the threshold by 2 – 3.5 ms, so the non-RTOS configuration should be expected to deliver the post-cliff push envelope (~51 N), not the nominal one. On a real-time OS with a dedicated balance-loop microcontroller (e.g., 1 kHz RT Linux or bare-metal MCU), jitter is typically <0.1 ms and 2 – 4 ms of budget is recoverable—which straddles the cliff rather than clearing it comfortably: 9.5 ms less 4 ms clears it, less 2 ms does not. Real-time scheduling is therefore a requirement for full push margin rather than an optimization, and the loop-latency target is 6 ms end-to-end, not 10 ms. A second route to the same end is to move the cliff instead of the budget: doubling the sagittal velocity damping shifts the knee to 8 – 10 ms, which the 9.5 ms estimate clears, at negligible cost to undelayed quiet stance (Table 12). That variant moves the push cliff only—the 14 – 16 ms fall threshold is unchanged—so it widens the performance margin without widening the safety one. We report it without adopting it (§5.6); a hardware build constrained to a non-RTOS budget is the case where re-qualifying it would be worth the cost. Neither scenario has been validated on hardware—latency characterization with an oscilloscope and jitter histogram is required before deployment, and it is the measurement that decides which side of the cliff a given build lands on. The discarded WBC-assist variant (§5.5) is excluded from the hardware path on control grounds, not compute grounds: its QP solve costs 0.169 ms mean and 0.30 ms p99 over 481 000 solves and would fit the budget, but admitting its torque degrades pitch by 2.5× and planar drift by 2.0×, while gating it to the point of harmlessness also gates it to 2.5 % of its permitted authority. Hardware degradation: 0.294 mm sagittal repeatability → projected 1 – 3 mm (encoder quantization, backlash, stiction, tire compliance, IMU noise; cf. Table 11). Staged commissioning: static torque → standing → push recovery → flight (tethered) → terrain.

We presented ACC, a two-channel torque assembly where a PD balance core is augmented by a proximity-gated anchor integral achieving 0.294±0.015 mm idle repeatability on the sagittal (actuated) axis—a 59× improvement over the proportional-only baseline under one unified protocol (N=10; Table 3)—with omnidirectional push survival (F_{min}=82 N, F_{med}=107 N). Absolute accuracy on that axis is reported separately and fully accounted for: the robot parks 27.0 mm from the commanded home and holds that standoff to 0.009 mm. A closed-form model (Appendix C) predicts the offset as the static price of a deliberately bounded integrator memory, matching the measured torque balance to 0.45 %, and both available corrections are priced against the evaluation suite. Six classical baselines—four 4-state TWIP variants (LQR, LQR+AW, LQR-DT, PI+AW) and two 6-state coupled variants (servo and direct-torque)—all fail (100 %); adding integral action to LQR-optimal feedback does not prevent the fall. Two control experiments test that comparison rather than assume it: sweeping the coupled model's one empirical constant over an 80× range moves the outcome not at all, because the roll channel is saturated throughout; and a full-state LQR derived from the plant itself—sound derivation, 0.54 % equilibrium residual, stabilizable, control weight swept over six decades—survives two orders of magnitude longer than the reduced-order designs and still falls, holding one commanded height at a time and recovering at most 20 N of push. ACC tolerates moderate noise, is insensitive to the actuator slew bound over an 8× span (100 – 800 Nm/s: 2 % spread in F_{max}, no resolvable change in lateral idle precision), and handles contact loss and terrain without neural network training. Its sharpest deployment constraint is actuator latency: push margin collapses across a cliff between 6 ms and 8 ms, below the 9.5 ms end-to-end budget we estimate for a non-real-time implementation. The contribution is thus a *structured classical prior*: an interpretable controller whose gating encodes the plant's measured failure modes, and whose intended downstream role is to supply the nominal base action of a residual-learning stack rather than to compete with a fully resourced learned policy. Future work: hardware validation, EMA settle-time characterization, a properly resourced learning comparison (multiple seeds, domain randomization, tuned reward), and extending conditional activation to locomotion and to residual RL over the prior.

Code and Data Availability

The controller, the MuJoCo model, every evaluation script referenced above, and the raw result files backing each table are available at `https://github.com/thuong180702/Wheeled-bipedal-robot-simulation` (branch `main`; a release tag will be minted at publication). Each main results table names its producing script and its raw result file in the note directly beneath it (Tables 3, **??**, **??**, 11 and 14); the remaining sweeps are cited inline at the point where they are discussed. The raw per-step records behind the whole-body-control results of §5.5—the 225-scenario three-arm campaign, the task-weight ablation, the gated and ungated assist runs, and the per-step gate trace—are collected under `outputs/wbc_postfix_evidence/`, whose `README.md` maps each file to the table it produces. The torque decomposition, gain sweep, height sweep, push-validation and noise/delay records behind Appendix C are collected the same way under `outputs/parking_offset_evidence/`; its `README.md` maps each file and each `--tag` to the table it produces, and gives the worktree recipe for reproducing a gain arm without disturbing the shipped profile.

Appendix A. Why Saturation-Triggered Anti-Windup Is Insufficient

Saturation-triggered AW triggers on actuator saturation; on this platform, position error grows to 10 – 30 cm during a push while wheels stay within their 20 rad/s range, so saturation-triggered AW never activates. A static dead-zone either over-restricts or under-restricts; ACC's proximity gate with continuous smoothstep avoids both. State-gated conditional integration is practiced in process control; ACC couples spatial gating with asymmetric envelope-detected damping, providing temporal gating absent in static-threshold schemes. AW-augmented baselines (LQR+AW, PI+AW) are evaluated in Table 8; broader AW taxonomy comparison is deferred.

Appendix B. WBC Baseline Implementation Details

2.1. Evidence Provenance

Every W1 and W2 number in §5.5 comes from a single campaign of the verified implementation (225 scenarios, 481 000 control steps); the full metric set is Table 10. The degraded-solver control experiment of §5.5 contributes exactly two quantities, both taken from the comparison report of that build (`docs/validation/V3_vs_V3_Assist_comparison_report.md`): QP success rate (85.85 %) and time-to-fall (0.52 s). No other quantity is drawn from it.

2.2. Scope of the Failure-Mechanism Claims

Exactly one failure mechanism is asserted for this baseline, and it is verified directly in the source: the posture task is a velocity damper, not a position tracker (§5.5). Two mechanisms that a reader might reasonably suspect are excluded by the implementation itself. The QP is not linearized at a fixed operating height—mass matrix, bias forces, CoM Jacobian and torso angular-velocity Jacobian are evaluated at the current state on every control step, and the module's only linearization is the standard, height-independent friction-cone pyramid. The wheels are not bare point contacts: lateral no-slip and forward rolling constraints are implemented and were active (`rolling_mode=full_rolling_soft`) throughout the campaign. The architectural claim—absence of regime-gating in a fixed-task-priority formulation—therefore rests on the invariance results of §5.5 (solver quality, task weighting), not on any deficiency specific to this implementation.

2.3. WBC Task-Weight Ablation

The standalone WBC's failure could in principle depend on the particular task weighting adopted. Table 15 reruns the WBC-only arm at four weightings over an identical scenario set. The spread between best and worst is 0.1 percentage points of fallen fraction and 0.03 s of upright time; all four collapse.

2.4. Assist Gate Decomposition

Table 16 shows why the gated assist is inert. The height factor $g_{\text{height}} = \exp[-((h - h_0)/\sigma_h)^2]$ uses h_0=0.53 m and σ_h=0.025 m, and is hard-zeroed below a 1×10^{-3} cutoff. Four of the five height variants in the scenario matrix command a base-z far enough from h_0 that the commanded-height term alone drives the gate shut before the measured height is even considered. On the one variant that remains open, the four state-dependent factors multiply to 0.461, and the per-joint agreement and role terms reduce the admitted authority by a further ~19×, to $\bar{\alpha}$=0.0061.

2.5. Baseline Implementation Verification

A baseline reported to fail must be shown to be correctly implemented, or its failure is evidence about the code rather than about the formulation. Four properties on which a task-priority QP-WBC depends were therefore checked against executable references before the campaign of §5.5 was run: the analytic contact Jacobian, the propagation of the task weighting into the QP cost, the frame of the height reference, and the frame of the assist authority gate. Table 17 states each check and its measured outcome. The checks are executable (`scripts/verify_wbc_audit_fixes.py`) and their raw output is archived at `outputs/wbc_audit_fix_verification.json`.

2.6. Why the Gated Assist Is Inert

Gate and controller agree on the height frame (V4), so the gate's near-total closure on this scenario matrix is a property of its *tuning*, not of a frame mismatch. With h_0=0.53 m and σ_h=0.025 m, four of the five height variants command a base-z that puts g_{height} below its $1{\times}10^{-3}$ cutoff on the commanded term alone (Table 16), and on the fifth the admitted authority is 2.5 % of $\alpha_{\max}$. The gated assist's numerical equivalence to ACC is therefore a direct consequence of a gate that is shut or nearly shut everywhere here, and is not by itself evidence that the blend is harmless. The evidence about the blend is the ungated configuration of §5.5, which admits WBC torque on 100 % of steps and degrades every stability metric but roll.

Appendix C. Static Torque Balance of the Sagittal Parking Offset

Section 5.1 reports a −27.0 mm sagittal parking offset on the axis the anchor integral regulates. This appendix derives the offset in closed form from a measured torque decomposition, validates the derivation against a 40× gain sweep and a five-point height sweep, and reports the measured cost of the available corrections.

All runs use the idle protocol of Table 3 (identical seeds, 5 s settle, 20 s window) with the controller's internal sagittal torque decomposition logged alongside the gravity moment about the wheel axles (`scripts/diagnose_parking_offset.py`; raw records under `outputs/parking_offset_evidence/`). Sagittal is world y on this platform; the x columns of the raw records are lateral.

3.1. *The offset is neither gravity-driven nor a saturated integral*

Two natural explanations fail on direct measurement (Fig. 7a, N=10, values averaged over the measurement window):

Gravity moment about the wheel axle	0.0040 Nm
CoM lever arm about the axle	0.050 mm
Anchor integral I	0.198 Nm (10.0 % of cap)
Integral clamp $I_{\max}$	2.0 Nm
Pitch-loop torque τ_{pitch}	−1.2587 Nm
Position-loop torque τ_{position}	1.2631 Nm
Adaptive bias trim τ_{trim}	0.0655 Nm
Controller position error e	−24.99 mm
Base parking offset $\bar{s}$	−26.96 mm

Gravity is not the load. The moment is 0.0040 Nm because the CoM sits 0.050 mm from the axle line. This is not a coincidence of tuning: a wheeled inverted pendulum that has stopped accelerating has necessarily brought its contact patch under its CoM, whatever its absolute position. The parking offset is a displacement of the whole machine relative to the commanded home, not a static lean, so no relation of the form $\tau_{\text{gravity}}(\bar{s}) \approx \tau_{\text{integral}}$ can hold.

The integral is not saturated. I reaches 10.0 % of its clamp during quiet stance and never approaches it, so the offset is not a windup or anti-windup artifact and cannot be relieved by raising the clamp. This is consistent with the ablation result of Table 3, where removing the proximity gate entirely (S4) reproduces ACC bit-for-bit.

What does balance is τ_{pitch} against τ_{position}, to within 0.005 Nm.

3.2. *Origin of the constant pitch torque*

The sagittal pitch term is $\tau_{\text{pitch}} = k_{p,\theta}(\theta - \theta_{\text{ref}})$ with $k_{p,\theta} = 50$ Nm/rad at idle. Decomposing the commanded reference at the nominal standing height h=0.4040 m:

Feedforward equilibrium-pitch calibration	+3.4963°
Support-position outer loop (PD, k_p=0.8145 °/m)	−0.0203°
Low-band support term (gated off at this height)	0.0000°
Commanded reference θ_{ref}	+3.4760°
Measured body pitch θ	+2.0336°
Residual $\theta - \theta_{\text{ref}}$	−1.4424°
$\Rightarrow \tau_{\text{pitch}}$	−1.2587 Nm

The decomposition closes to four decimal places. The offset therefore originates in a 1.44° calibration error of the feedforward equilibrium-pitch map at this operating point, not in the anchor at all. Note that the outer loop, which does observe the 25 mm error, contributes 0.02° against the 1.44° it would need: its integral channel is disabled, so it too has finite DC gain.

3.3. *Closed form*

Writing the sagittal wheel torque during quiet stance as $\tau_{\text{pitch}} + \tau_{\text{position}}$, with the position law unsaturated ($|\tau_{\text{position}}| \approx 1.3$ Nm against a 4.0 Nm cap),

$$\tau_{\text{position}} = -k_{pos}\, e + I + \tau_{\text{trim}}. \tag{24}$$

The implemented anchor integral is *leaky*—Eq. (9) writes the ideal integral for clarity, but the discrete update applied at every step is

$$I_{k+1} = (1-\lambda)\big(I_k - k_i\,\Delta t\, g_{\text{prox}} g_{\text{env}}\, e_k\big), \tag{25}$$

with λ=5×10^{-3} per step (Table 2). A leaky integrator is a first-order lag, not an integrator: at a fixed point with the gates open, Eq. (25) gives $I^* = -k_{I,\text{dc}}\, e$ with

$$k_{I,\text{dc}} = k_i\,\Delta t\,\frac{1-\lambda}{\lambda} = 4.0 \cdot 0.01 \cdot \frac{0.995}{0.005} = 7.96\ \text{Nm/m}, \tag{26}$$

which is one fifth of k_{pos}=40 Nm/m, not infinity. Substituting into Eq. (24) and imposing zero net sagittal torque:

$$\begin{aligned} e_{ss} &= \frac{|\tau_{\text{pitch}}| - \tau_{\text{trim}}}{k_{pos} + k_i\,\Delta t\,(1-\lambda)/\lambda} \\ &= \frac{1.2587 - 0.0655}{40 + 7.96} = 24.88\ \text{mm}, \end{aligned} \tag{27}$$

against 24.99 mm measured—an error of 0.45 %. The remaining 2.0 mm between e and the base offset $\bar{s}$ is accounted for by the base-frame versus support-centre convention gap, so the two figures are consistent once expressed in the same frame.

The conclusion is structural rather than incidental: *no element of ACC's sagittal position chain has infinite DC gain.* The anchor integral is leaky by design and the outer loop's integral channel is disabled, so a constant disturbance torque must be paid for with a constant position error. The 27 mm is the price of that design choice, in millimetres per newton-metre.

3.4. Validation across a forty-fold gain sweep

Sweeping k_i from 4 to 160 at fixed leak moves $k_{I,\mathrm{dc}}$ over 40× and the offset follows Eq. (27) throughout (Fig. 7b):

k_i	$k_{I,\mathrm{dc}}$ (Nm/m)	e meas. (mm)	e pred. (mm)	window jitter (mm)
4	**7.96**	**-24.99**	**-24.87**	**0.298**
10	19.90	-20.42	-20.34	0.302
20	39.80	-15.49	-15.54	0.147
40	79.60	-10.57	-10.67	0.208
60	119.40	-7.96	-8.01	0.168
80	159.20	-6.40	-6.41	0.094
120	238.80	-4.59	-4.58	0.114
160	318.40	-3.56	-3.56	0.066

Steadiness improves monotonically with k_i over this range rather than degrading, which is the opposite of the integral/steadiness trade-off measured at k_i=0 in Table 3 and indicates that the 3× steadiness cost reported there is paid at the first newton-metre of integral authority, not progressively. Every entry above is measured on clean sensing with no actuator delay; §3.5 shows that this is exactly the condition under which raising k_i looks free.

3.5. Why the gain is not raised: the noise–delay cost

The sweep above and the push protocol below both make k_i=160 look free: the offset falls by 79.5 %, clean-idle jitter improves, and neither $F_{\min}$ nor F_{med} moves. Those three probes share a blind spot—all are run on clean sensing, and the push probe is a large transient rather than a sustained small one. Re-running the two cells of Table 11 that combine medium sensor noise with actuator delay, at N=20 per cell and identical seeds across arms, prices the change:

k_i	$k_{I,\mathrm{dc}}$ (Nm/m)	$\bar{s}$ (mm)	offset removed	falls (of 40)	Fisher vs. k_i=4
4	**7.96**	**-26.96**	**—**	**7**	**—**
20	39.80	-17.47	35.2 %	13	p=0.196
40	79.60	-12.55	53.5 %	10	p=0.586
80	159.20	-8.37	68.9 %	14	p=0.126
160	318.40	-5.53	79.5 %	17	p=0.027

A Cochran–Armitage trend test on $\log_2 k_i$ gives z=2.34, p=0.019: the fall rate rises with integral gain across the sweep, and pooling every arm above the shipped setting gives 54 falls in 160 against 7 in 40 (p=0.055). Because the cost is graded rather than a threshold, selecting the largest gain that is not individually significant would be an artifact of sample size—k_i=40 carries p=0.59 only because 40 trials cannot separate 17.5 % from 25 %. Zero-delay cells stay clean at every gain, and the idle, push, flight, terrain and rate-limit metrics are flat across the sweep, so the regression is specific to noise and delay *together*. One trap is worth naming: the idle column of Table 11 averages survivors only, so in these two cells the higher-gain arms report *better* idle numbers while falling more often.

The mechanism follows from Eq. (25). The leaky integrator has transfer function $k_i\Delta t/(\lambda + s\Delta t)$: a DC gain of $k_i\Delta t/\lambda$ and a corner at $\lambda/\Delta t \approx 0.5\,\mathrm{rad/s}$. Raising k_i scales the response at *every* frequency, so it lifts mid-band gain—with the 90° lag of an integrator—in precisely the band where the 10 ms actuator delay of those cells has already consumed the phase margin. Lowering λ raises DC gain alone, leaving mid-band untouched, but costs the memory horizon instead (next subsection). Neither single parameter separates the two roles; doing so requires a lead-compensated or explicitly bandwidth-limited integrator, which is a structural change to a controller whose present form is qualified over a 48-scenario suite, and is left as future work.

3.6. The leak sets memory, not accuracy

Because $k_{I,\mathrm{dc}}$ depends on both k_i and λ, reducing the leak also raises DC gain—but it is the wrong knob. Cutting λ from 5×10^{-3} to 5×10^{-4} raises $k_{I,\mathrm{dc}}$ to $79.96\,\mathrm{Nm/m}$ and does reduce the offset to -16.31 mm, but window jitter degrades from 0.298 mm to 3.463 mm, a factor of 12, and the measured error (-14.34 mm) does not reach its DC prediction (-10.41 mm) because the forgetting time constant τ=$\Delta t/\lambda$ has grown to 20 s and no longer settles inside the measurement window. Since τ is independent of k_i, the two parameters separate cleanly: k_i sets DC gain, λ sets the memory horizon. The shipped pair is chosen on that basis. λ=5×10^{-3} fixes a 2 s forgetting time, which is what keeps the integral inside its measurement window; k_i=4 is then set not for clean-idle steadiness, which the gain sweep shows is flat in k_i, but for margin against the noise-plus-delay conjunction quantified in §3.5. The accuracy consequence of that pair is the 27 mm offset, and it is the price of the memory horizon rather than of the gain.

3.7. Two measured corrections and their cost

The height dependence of the calibration error was checked on the five posture-calibrated height variants, applying the correction measured at each point. The nominal row is N=10 under the full idle protocol; the four off-nominal heights are single trials, whose run-to-run spread the nominal row bounds at 0.011 mm SD:

h (m)	correction (°)	$\bar{s}$ before (mm)	$\bar{s}$ after (mm)	reduction
0.394	−1.622	−28.77	−1.16	96.0 %
0.399	−1.974	−34.53	−1.20	96.5 %
0.404	**−1.461**	**−26.96**	**−2.02**	**92.5** %
0.409	−0.943	−19.29	−2.83	85.3 %
0.414	−1.295	−25.11	−2.92	88.4 %

The controller's own position error falls to ≤0.12 mm at every height and the anchor integral to 0.4 % of its clamp; what remains is the 1 – 3 mm frame-convention gap. The required correction is not a constant (it spans 0.94 – 1.97° over 2 cm), so this is a per-height recalibration of the feedforward map rather than a single scalar.

Both corrections were then put through the full push protocol of §5.2 (N=10 repetitions × 8 bearings, binary search over 10 – 160 N at 5 N tolerance; Welch's t against the published arm):

Arm	F_{min} (N)	F_{med} (N)
ACC as published (k_i=4)	82.8(10)	116.8(47)
k_i=40	81.8(18) p=0.14	116.8(70) p=0.98
k_i=160	81.9(27) p=0.32	115.3(36) p=0.43
Feedforward pitch recalibrated	82.5(12) p=0.55	112.4(49) p=0.05

The published arm reproduces the weakest bearing of §5.2 (82.8 N here against 82.3 N there, Δ<0.5 N on independent seeds), validating the harness. Raising k_i by 40× costs nothing measurable on either push statistic and removes 79.5 % of the offset; the feedforward recalibration removes 92.5 % and leaves F_{min} unchanged, but shows a 4.5 N (3.8 %) reduction in F_{med} at p=0.051, which at this sample size we report as unresolved rather than as null.

Neither correction is promoted into the shipped profile, and in the k_i case the reason is not evidence-base consistency but a measured regression: the push protocol is blind to the failure mode that gain change introduces, which appears only when sustained sensor noise and actuator delay act together (§3.5). This is the substantive result of the appendix. The 27 mm offset is not an unexamined residual, nor a defect awaiting a constant change; it is the position error that a deliberately leaky, deliberately low-gain integrator must pay in exchange for a 2 s forgetting horizon and for phase margin against delay. Both alternatives were measured across the full evaluation suite rather than on the accuracy metric alone, which is what exposed the price. The feedforward recalibration remains the more promising route, since its cost is confined to one borderline statistic; resolving that at a larger sample, and testing whether a bandwidth-limited integrator can raise DC gain without raising mid-band gain, are the two concrete steps toward closing this limitation.

Notes on contributor

Van Thuong Nguyen is an independent researcher. His research interests include classical and learning-based control for wheeled bipedal robots, with a focus on structured controllers used as priors for residual reinforcement learning.

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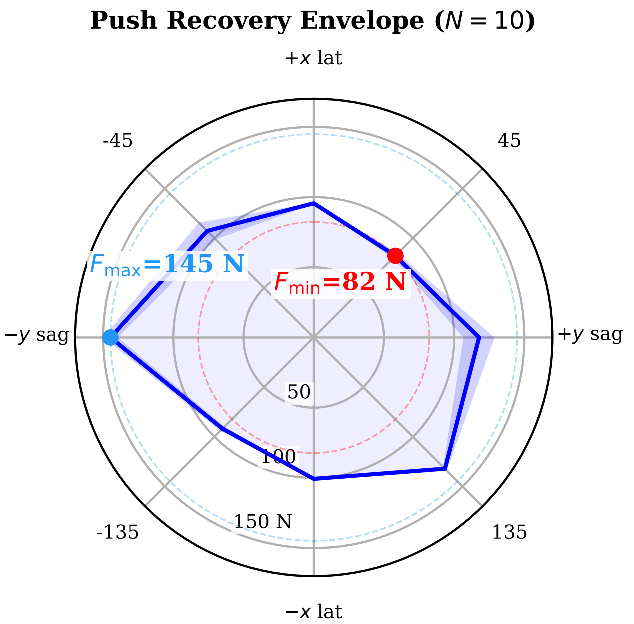

Figure 4. Omnidirectional push envelope of ACC. This is the *same* run as row S1 of Table **??**, not a separate measurement: 8 bearings, N=10 repetitions each, threshold located by binary search over 10 – 160 N to a 5 N tolerance, so the plotted minimum is by construction the $F_{\min}$ of that row. Solid curve: per-bearing mean survivable push magnitude; shaded band: ±1 standard deviation over the ten repetitions. The radial axis is linear in force and originates at 0 N, so radial distance is proportional to margin. The envelope is strongly anisotropic, spanning 82.3 – 144.8 N—a 1.8× ratio between the weakest and strongest bearing. The ordering is −90° (144.8 N) > 135° (132.2 N) > 90° (117.8 N) > −45° (107.3 N) > 180° (100.7 N) > 0° (95.6 N) > −135° (91.9 N) > 45° (82.3 N). Two regularities are visible. First, the actuated axis is the strong one: bearings are applied in the world frame as $\mathbf{f} = F(\cos\theta, \sin\theta, 0)$, so ±90° are purely sagittal and 0°/180° purely lateral (Sec. 5.1). The two sagittal bearings average 131.3 N against 98.2 N for the two lateral ones, a 1.34× advantage. This is the expected asymmetry for a wheeled biped: sagittally the wheels accelerate the contact patch back under the CoM and convert the impulse into a controlled excursion, whereas laterally no actuator produces ground force at all and the only restoring authority is the static $w_{\text{track}} = 0.278$ m track width plus hip-roll PD. We compare axis *means* rather than ranks because the raw ordering is contaminated by the second regularity. Second, the envelope is measurably *chiral*: 135° survives 40.3 N more than −135°, and −90° 27.0 N more than 90°. This is not sampling noise and not the plant, which is mirror-symmetric to numerical precision: it is the sign convention of the controller's antisymmetric hip-roll and hip-yaw channels, which leaves the closed loop C_2-equivariant but mirror-broken. The text traces it through the sin 2θ harmonic and closes the argument with two control experiments. $F_{\min}$ (red) is the bearing-wise minimum and is the figure of merit quoted throughout the paper, because it is the magnitude survived in *every* direction; $F_{\max}$ (blue) is shown only to indicate the spread.

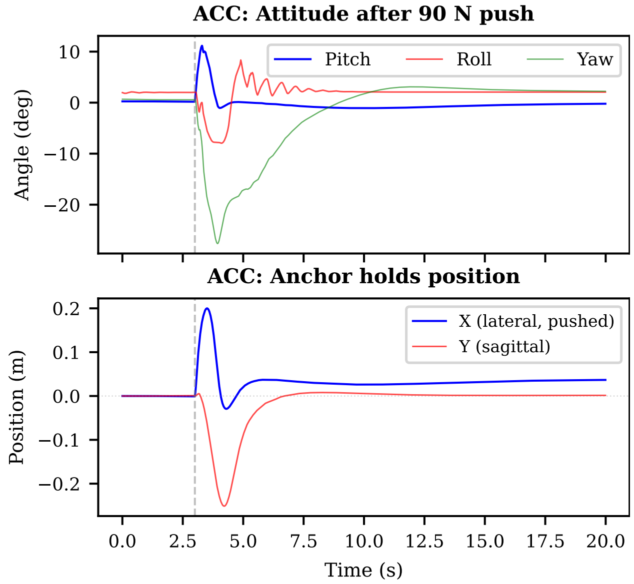

Figure 5. Post-push recovery (90 N lateral, world +x, at t=3 s). **Top:** attitude angles; stabilization within 2 s. **Bottom:** planar CoM position, which separates the two axes cleanly. The lateral push couples through roll into a large *sagittal* excursion (−251 mm peak); the anchor drives that axis back to 1.37 mm of home and holds it there at 0.018 mm SD over the final 2 s—a 99.5 % recovery. On the pushed lateral axis, which has no actuator authority (Sec. 5.1), the 200 mm peak excursion decays only to a permanent 36.2 mm offset, 18 % of the peak. The contrast between the two traces is the clearest single illustration of what the anchor does and where it does not act.

Table 5. Oblique ledge descent: settle count out of 8 independent initial conditions, at 0.7 m/s. Two regimes separated by a dead band, not a height threshold. Bold marks the dead band at each angle.

Step	30°	45°	Step	30°	45°
12 cm	8/8	1/8	27 cm	**0/8**	5/8
15 cm	8/8	**0/8**	30 cm	1/8	4/8
17 cm	6/8	**0/8**	32 cm	3/8	**0/8**
20 cm	2/8	**0/8**	35 cm	4/8	5/8
22 cm	**0/8**	**0/8**	37 cm	6/8	6/8
25 cm	**0/8**	**0/8**	40 cm	5/8	3/8
Total: 30° 43/96; 45° 24/96					

Survival is the strict settle verdict of Table **??**. The 32 cm/45° cell is a reproducible 0/8 between 50 % and 62 % neighbours and is reported unsmoothed. Produced by `scripts/ramp_step_tests.py--seeds`; raw output `outputs/oblique_regime/diag\{30,45\}_fine.txt`.

One-wheel curb (20 cm) — per-leg terrain adaptation

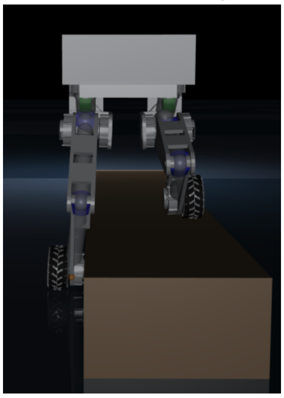

Figure 6. One-wheel curb (20 cm). With the per-leg height split: 14.9±0.1° straddle roll, 10/10 settled. With a symmetric posture (split removed): the curb-side leg cannot fold far enough, roll diverges past 60° within 0.8 s of curb entry, 0/10.

Table 7. Baseline gain derivation. All weights follow Bryson's rule (Bryson & Ho, 1975); see the note below the table.

Controller	Gains and derivation
LQR (4-state)	$Q=\mathrm{diag}(10, 2, 3, 0.3)$, $R=0.8$; roll and yaw closed by decoupled PD through the PID servo layer
LQR + AW	as above, plus a pitch-error integral with conditional anti-windup, clamped to ±3.0 rad/s
LQR, DT	4-state gains re-optimized for the zero-lag plant: $\tau_s=0$, $k_{p,\mathrm{roll}}=55.0$
6-state LQR	$Q=\mathrm{diag}(10, 2, 3, 0.5, 3, 0.3)$, $R=\mathrm{diag}(0.8, 1.0)$, derived in continuous time and shared by the servo and DT variants
6-state LQR, DT	state feedback re-solved for the zero-lag plant: $K_{\mathrm{pitch}}=113$, $K_{\mathrm{roll}}=58$ Nm/rad (see Limitation 4)
PI + AW, DT	DT-LQR gains plus an integral on forward position: $k_{i,\mathrm{pos}}=8.0\,\mathrm{rad/s/(m{\cdot}s)}$, clamp ±5.0 rad/s, conditional gate ±15 cm
PID servo layer	$k_p=[55, 40, 70, 70, 4]$, $k_d=[3, 2, 4, 4, 0]$, $k_i=[0.8, 0.4, 1, 1, 0.1]$, ordered (hip roll, hip yaw, hip pitch, knee, wheel)
Servo lag	0.25 s, measured as the wheel-velocity 90 % rise time (≈ 25 steps at 100 Hz)

Bryson's rule throughout ($Q_{ii}=1/x_{i,\max}^2$, $R=1/u_{\max}^2$); the 46-configuration retuning sweep described in the text confirmed that no retuning within these ranges changes the outcome. The 4-state model is (pitch, pitch-rate, forward velocity, forward position); the 6-state model adds (roll, roll-rate). "DT" denotes direct torque, i.e. the PID servo path removed and $\tau_s=0$.

Table 8. Classical baselines and the preliminary PPO reference vs. ACC, clean sensors. All rows at 100 Hz control except the PPO reference, which is reported at the 50 Hz rate its policy was trained at. Parenthesised N is the episode count. See the note below the table.

Configuration	Surv. (s)	θ_{RMS} (°)	ϕ_{RMS} (°)	H_{RMS} (mm)	τ_{RMS} (Nm)
LQR, TWIP (N=20)	0.32	0.006	24.2	205.8	21.99
LQR + AW (N=20)	0.32	0.006	24.2	205.8	21.99
LQR, DT (re-opt.) (N=20)	0.60	0.033	27.5	119.2	8.76
6-St. LQR (N=20)	0.32	0.006	24.2	205.8	21.99
PI + AW, DT (N=20)	0.52	0.181	26.1	171.3	10.42
6-St. LQR, DT (N=20)	0.16	0.001	21.2	128.5	14.44
PPO ref. (50 Hz) (N=20)	3.20	4.82	3.15	54.2	18.40
ACC (N=10)	**20.00**	**0.060**	**0.010**	**0.18**	**3.28**

LQR / LQR+AW / 6-St. use the PID servo; LQR (DT), PI+AW (DT) and 6-St. (DT) command torque directly; PPO is a 5.4M-step model-free run (preliminary, single seed, no domain randomization), reported as a reference point rather than a tuned RL baseline. The identical LQR / LQR+AW / 6-St. rows arise because all three command through the same PID servo path, whose response dominates before the gain differential between them can act; the coincidence holds at both control rates (Table **??**).

Between-episode standard deviations (omitted above for legibility): survival time is deterministic on all six classical rows (±0.00 s at N=20, seeds $\{0, 42, 123\}$); PPO: ±0.85, ±0.91, ±0.45, ±8.3, ±2.1; ACC: ±0.00, ±0.006, ±0.001, ±0.02, ±0.00. Per-metric between-episode dispersions for the classical rows are not retained in the rate-matched output (`outputs/rate_matched_baselines/results.json`), which stores episode-aggregated values.

All LQR and PI+AW variants fall within 0.60 s (100 %); the 6-St. DT variant falls deterministically at 0.16 s (zero trial-to-trial variance), confirming that removing the servo path with gains derived from a simplified model does not rescue stability. PI+AW (DT-LQR + integral AW on fwd. position) does not improve on DT-LQR at this rate—it is 0.08 s worse—because the integral cannot accumulate usefully before the roll-induced fall. The PPO reference achieves partial standing but suffers 85 % falls over multi-height transitions (H_{RMS}=54.2 mm). N=20 throughout; classical rows from `outputs/rate_matched_baselines/results.json`.

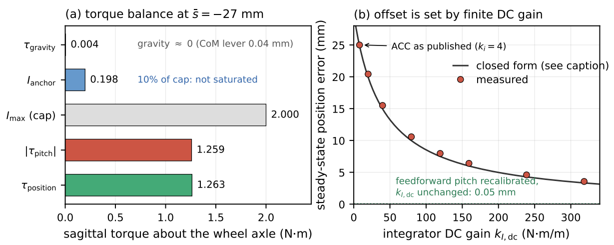

Figure 7. Static torque balance of the sagittal parking offset. (a) Measured sagittal torques at the $\bar{s}=-27$ mm parking point. The gravity moment about the wheel axle is 0.0040 Nm and the anchor integral sits at 10 % of its clamp, so the offset is neither gravity-driven nor a saturation artifact; the balance is between the pitch and position loops. (b) Steady-state position error against integrator DC gain $k_{I,\mathrm{dc}}$ of Eq. (26), swept over $k_i \in [4, 160]$ at fixed leak. The curve is Eq. (27) evaluated with the baseline τ_{pitch}; using each run's own measured τ_{pitch} the closed form tracks all twelve gain and height points to within 0.92 %. The dotted line is the same controller with the feedforward pitch reference recalibrated and $k_{I,\mathrm{dc}}$ left unchanged.

Table 10. Three-arm counterfactual under cloned initial conditions: ACC alone, standalone QP-WBC, and WBC blended as an assist on ACC with the authority gate disabled and enabled. All arms use the verified WBC implementation (Appendix B).

Metric	ACC	WBC	ACC + WBC assist: gate off	ACC + WBC assist: gate on
Full batch: 225 scenarios, 481 000 control steps				
Scenarios with a fall	0/225	225/225	0/225	—
Episode fallen (%)	0.0	97.3	0.0	—
Upright time (s)	full	0.570	full	—
Pitch RMS (°)	0.52	2.33	1.295	—
Roll RMS (°)	3.71	93.45	**3.20**	—
Roll max (°)	8.37	103.89	7.64	—
Yaw drift (°)	6.69	6.80	12.56	—
Base height RMS (m)	0.540	0.126	0.518	—
Final height (m)	0.538	0.091	0.515	—
Planar drift (m)	0.099	0.165	0.199	—
Torque RMS (Nm)	3.18	0.96	3.67	—
Max abs. torque (Nm)	9.70	4.33	10.74	—
QP success (%)	—	99.99	100.00	—
Gate-open subset: 5 scenarios, 10 775 steps				
Pitch RMS (°)	0.520	—	1.284	0.519
Roll RMS (°)	3.545	—	**2.903**	3.540
Yaw drift (°)	5.479	—	6.533	5.463
Base height RMS (m)	0.5412	—	0.5190	0.5412
Planar drift (m)	0.0674	—	0.1884	0.0677
Torque RMS (Nm)	3.154	—	3.664	3.154

"Full" upright time denotes no fall in any episode (20 s for fixed-height and height-transition scenarios, 21.55 s for push scenarios). Base height RMS is the root-mean-square of *absolute* base-z, so a larger value means the robot stayed up: WBC's 0.126 m is a collapsed torso, not tighter tracking. Torque RMS is likewise not an efficiency measure here—WBC's 0.96 Nm is the torque of a machine lying on the ground. Roll is the sole metric on which the assist beats ACC (bold); see §5.5. The gate-open subset is the `low_small` height variant, the only one of five on which g_{height} does not collapse (Table 16); the gated arm was not run on the other four because its gate is identically closed there.

Table 11. Robustness to sensor noise and actuator delay, N=20 per cell. Falls: trials terminated by a fall, out of N. Idle RMS: lateral (world x) CoM std. over 20 s after 3 s settle (EMA_0=0 transient), *surviving trials only*. F_{max}: binary-search lateral (world $+x$) push, ±5 N tolerance, search floor 10 N. Delay is quantised to the control period here; the axis is resolved five times more finely in Table 12.

Condition	Falls	Idle RMS (mm): Mean	Idle RMS (mm): ±Std	F_{max} (N): Mean	F_{max} (N): ±Std
Clean, 0 ms delay	0/20	0.43	0.06	95.7	0.8
Clean, 10 ms delay	0/20	2.43	0.21	50.5	7.9
Clean, 30 ms delay	**20/20**	— (all fell)		—	
Low noise, 0 ms delay	0/20	5.86	2.53	96.0	1.0
Low noise, 10 ms delay	0/20	10.66	3.70	51.3	8.9
Low noise, 30 ms delay	**20/20**	— (all fell)		—	
Med noise, 0 ms delay	0/20	30.82	12.13	93.5	5.2
Med noise, 10 ms delay	**2/20**	35.38	21.63	51.3	13.4
Med noise, 30 ms delay	**20/20**	— (all fell)		—	
High noise, 0 ms delay	20/20	— (all fell)		—	
High noise, 10 ms delay	20/20	— (all fell)		—	
High noise, 30 ms delay	20/20	— (all fell)		—	

Noise (1-σ): low = 0.01 rad/s (ang. vel.), 0.01 m/s^2 (grav.), 0.001 rad (joint pos.); med = 5× low; high = 10× low. Delay is applied as a transport lag on the actuator command: the torque written n control periods ago is the one the plant receives. High noise is unrecoverable at every delay including zero (60 of 60 trials): it holds the velocity envelope above threshold, disengaging the anchor, and the residual P-only mode cannot hold the platform under that noise level. Its failure is therefore a noise failure, not a delay failure. Because the idle column averages survivors only, the medium-noise 10 ms row is mildly optimistic; the falls column is the primary metric for that cell. Produced by `scripts/collect_robustness_sweep.py`, raw output `outputs/paper_statistics/robustness_sweep.json`.

Table 12. Actuator-delay resolution sweep, clean sensing. Transport delay applied per physics substep (2 ms at 500 Hz) rather than per control step, giving 2 ms resolution. F_{max}: binary-search lateral push, 7 iterations over 10–160 N. ACC: promoted default. ACC-DH: delay-hardened variant, sagittal velocity damping 22.5→45 Nm/(m/s), reported but *not* adopted. Clean sensing is deterministic (σ=0.0 in every multi-seed cell, N up to 5), so single-seed cells are exact.

Delay (ms)	ACC F_{max} (N)	ACC-DH F_{max} (N)	Change
0	95.5	95.5	—
2	94.4	94.4	—
4	93.2	94.4	+1.3%
6	93.2	94.4	+1.3%
8	**51.0**	**93.2**	**+83%**
10	56.9	60.4	+6.2%
12	41.6	79.1	+90%
14	40.5	68.6	+69%
16	28.8	41.6	+44%
20	both at search floor (10 N)		—

The shipped controller's knee lies between 6 ms and 8 ms: flat to within 2.4 % out to 6 ms, then a 45 % collapse, declining monotonically thereafter to no recoverable push by 20 ms. ACC-DH moves the knee to 8–10 ms and is no worse at any delay tested. Substep granularity bounds the resolution at 2 ms; neither knee is localized further.

The 10 ms cell is the only pure one-control-period hold; the others mix two consecutive commands within a period, which is why F_{max}

Table 14. Compound disturbance: lateral (world $+x$) push delivered at the midpoint of a 1 s commanded height ramp Δh about the 0.404 m nominal CoM-z. N=10 per row, independent initial-posture draws, binary search over [10, 130] N. Δh=0 is the matched static control. Rows within ±5 cm stay inside the calibrated posture band (0.354 – 0.454 m); the ±10 cm rows are extrapolated. The right-hand column is a control arm in which the same posture family is re-parameterized so that the achieved CoM height equals the commanded one.

		Shipped posture map			Re-calib.
Δh	Band	F_{max} (N)	vs. static	Pitch (°)	F_{max} (N)
+10 cm	extrap.	29.7 ± 8.7	−68.7 %	12.6 ± 13.1	52.8 ± 12.1
+5 cm	calib.	89.9 ± 0.9	−5.1 %	12.7 ± 0.4	89.7 ± 0.9
0 (static)	—	**94.8 ± 0.8**	—	14.0 ± 0.2	**94.8 ± 0.8**
−5 cm	calib.	101.0 ± 1.0	+6.5 %	18.9 ± 3.2	100.6 ± 0.9
−10 cm	extrap.	94.8 ± 1.9	±0.0 %	28.9 ± 1.8	93.2 ± 2.0

Mean±SD. Against the static control, all rows are significant under Welch's t-test at the Bonferroni-adjusted α=0.007 of §6 (p<3×10^{-9}) *except* −10 cm, which is indistinguishable from it (p=0.99). In the re-calibrated arm only the +10 cm row differs from the shipped map (p=1.5×10^{-4}); the other three are within noise (p≥0.10), as they must be, since the two maps agree to 3.2 mm inside the calibrated band. Peak pitch is measured over the whole episode at each trial's own F_{max}, so it is a shape statistic at the survival boundary, not a severity ranking across rows. Produced by `scripts/compound_disturbance_corrected.py`; raw outputs `outputs/compound_disturbance/results_corrected_map_shipped.json` and `..._map_comcalib.json`. Posture-map accuracy is verified by `--self-check` against the model's own forward kinematics.

Table 15. WBC-only arm at four task weightings (`random_push`, nominal height, seeds 201–205, identical scenarios). No weighting rescues the baseline.

Task weighting	Fallen (%)	Upright (s)	Roll RMS (°)	Pitch RMS (°)
Balanced (default)	97.4	0.570	92.86	2.51
Torso priority	**97.3**	0.590	95.28	**1.52**
Posture priority	97.4	0.570	93.52	2.37
CoM priority	97.4	0.560	93.15	2.42
ACC, same scenarios	**0.0**	full	**3.71**	**0.52**

Mode names are `balanced_default`, `torso_priority`, `posture_priority`, `com_priority` in `wheeled_biped/wbc/offline_task_stack.py`. The balanced default is w_{torso}=3, w_{com}=3, $w_{posture}$=1.5; each priority mode raises its named weight to 10. Torso priority gives the best pitch and the worst roll: torso attitude is a single SO(3) task carrying equal roll and pitch gains, so raising its weight redistributes error between the two axes instead of reducing both.

Table 16. Assist-gate decomposition. Top: the commanded-height term across the scenario matrix. Bottom: measured per-step gate factors on the one variant where the gate opens.

Height variant	Commanded base-z (m)	g_{height}
`high_small`	0.750	2.3×10^{-34} → 0
`high_tiny`	0.670	2.4×10^{-14} → 0
`nominal`	0.650	9.9×10^{-11} → 0
`low_tiny`	0.630	1.1×10^{-7} → 0
`low_small`	0.550	**0.53** (open)
Measured on `low_small`, 10 775 control steps		
g_{height}		0.527
$g_{stability}$		0.876
g_{push}		0.998
$g_{divergence}$		0.999
product of the four		0.461
$\bar{\alpha}$ admitted		**0.0061** (max 0.0146)
$\bar{\alpha}/\alpha_{max}$		2.5 %

α_{max}=0.25. The remaining shortfall between the 0.461 factor product and the admitted 0.0061 is the per-joint agreement term A_j and role weight $K_{role,j}$: the WBC's torque opposes ACC's on most joints, and disagreement attenuates the blend. h_0 and σ_h are hand-tuned constants of *ACC's* gate (σ_h was already widened from 0.015 m); they bound the authority ACC grants the WBC and are not a property of the WBC formulation, which is state-dependent and carries no fixed validity band.

Table 17. Verification of the WBC baseline implementation. Each row is an executable check run against the implementation used for every result in §5.5; all four pass.

ID	Check	Measured result
V1	Analytic contact-point translational Jacobian compared element-wise against MuJoCo `mj_jac` at both wheel contact points; leg-column norms inspected for zero-gradient degeneracy.	Max abs. error 4.2×10^{-8} (full), 3.4×10^{-8} (actuated block), against an acceptance bound of 1×10^{-3}. Ipsilateral leg columns nonzero (norms 0.042, 0.433, 0.403, 0.306 and 0.060); contralateral columns exactly zero, as the kinematics require.
V2	Task mode must reach the QP cost, i.e. the cached cost builder must be parameterized by `task_mode`.	Builder takes `task_mode`, so the four weightings of Table 15 are genuinely distinct problems; the one call site passing a literal is the task-independent constraint-only fallback, which carries no cost terms by construction.
V3	The batch runner must pass CoM-z, not base-z, as the ACC height reference (§3.1); substituting one for the other injects a 13.2 cm reference error.	Height reference is read from the scenario's `target_com_z`; no base-z path into the height command exists in the source.
V4	The adaptive-assist g_{height} gate must compare against its own base-z target rather than reusing the CoM-z reference.	Gate and controller agree on the frame: a separate base-z target is passed. The gate is nonetheless nearly shut on this scenario matrix for the tuning reason quantified below.

Checks are executable: `scripts/verify_wbc_audit_fixes.py`; archived output `outputs/wbc_audit_fix_verification.json`.